MILITARY SPECIFICATION

CARRIAGE, 155MM HOWITZER: M39

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of lightweight carriage designed for use with the 155MM Howitzer, M198.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530	-Plywood, Flat Panel
QQ-S-781	-Strapping, Steel, and Seals
CCC-C-429	-Cloth, Osnaburg, Cotton
PPP-P-291	-Paperboard, Wrapping, and Cushioning
PPP-T-42	-Tape, Packaging/Masking, Paper
PPP-T-60	-Tape, Packaging, Waterproof

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MIL-P-116	-Preservation-Packaging, Methods of
MIL-B-121	-Barrier Material, Greaseproofed, Waterproofed, Flexible

FSC 1025

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-130	-Paper, Wrapping, Laminated and Creped
MIL-C-5501	-Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-C-6021	-Castings, Classification and Inspection for
MIL-H-6088	-Heat Treatment of Aluminum Alloys
MIL-I-6866	-Inspection, Penetrant Method of
MIL-I-6868	-Inspection Process, Magnetic Particle
MIL-S-6872	-Soldering Process, General Specification for
MIL-H-6875	-Heat Treatment of Steels (Aerospace Practice, Process for)
MIL-S-7742	-Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-R-11468	-Radiographic Inspections, Soundness Requirements for Arc and Gas Welds in Steel
MIL-P-11432	-Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-E-16053	-Electrodes, Welding, Bare, Aluminum Alloys
MIL-C-16555	-Coating Compound, Strippable, Sprayable
MIL-W-45205	-Welding, Gas Metal - Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-I-45607	-Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-R-45774	-Radiographic Inspection, Soundness Requirements for Fusion Welds in Aluminum and Magnesium Missile Components
MIL-W-63150	-Weapons and Support Material, Std Quality Assurance Provisions, for

STANDARDS

FEDERAL

FED STD. NO. 595	-Colors

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MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-130	-Identification Marking of U.S. Military Property
MIL-STD-171	-Finishing of Metal and Wood Surfaces
MIL-STD-248	-Welding and Brazing Procedure and Performance Qualification
MIL-STD-00453 (USAF)	-Inspection, Radiographic
MIL-STD-1252	-Inertia Friction Welding Process, Procedure and Performance Qualification
MIL-STD-1261	-Welding Procedures for Constructional Steels
ATAC-STD-113	-Reference Standards and Radiographic Procedures for Partial Penetration Aluminum Welds.
ATAC-STD-114	-Reference Standards and Radiographic Procedures for Partial Penetration Steel Welds.

DRAWINGS

AMCCOM

K12007600	-Carriage Assembly, 155MM Howitzer, M39

(Product drawings referenced in this specification form a part of Drawing K12007600).

PUBLICATIONS

ARRADCOM

LQ12007600	-Quality Requirements List, Carriage, 155MM Howitzer, M39
EL12007600-I	-List of Inspection Equipment Numbers, Carriage, 155MM Howitzer, M39

U.S. ARMY

AR702-3	-Army Materiel Reliability, Availability and Maintainability (RAM)

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	-Surface Texture
ANSI Y14.5	-Dimensioning and Tolerancing

(Applications for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	-Symbols for Welding and Nondestructive Testing

(Application for copies should be addressed to: The American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 Standards for Federal Service	-Screw-Thread

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

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3.2 Materials, construction and design. The carriage shall conform to the materials, construction and design requirements specified herein and on Drawing K12007600.

3.2.1 Heat treatment. Unless otherwise specified on the applicable drawings, heat treatment for aluminum alloys shall conform to MIL-H-6088, and heat treatment for steel alloys shall conform to MIL-H-6875.

3.2.2 Dimensions. Dimensions and tolerancers of parts and assemblies of the carriage shall conform to the detailed drawings of Drawing K12007600. Interpretation of the dimensions and tolerances shall be in accordance with ANSI Y14.5.

3.2.3 Surface texture. Surface texture shall be as specified on the applicable drawings. Interpretation of surface roughness symbols shall be as defined in ANSI B46.1.

3.2.4 Protective finishes.

3.2.4.1 Inorganic finishes. Parts requiring an inorganic finish shall be finished or coated as indicated on the applicable drawings and in accordance with MIL-STD-171.

3.2.4.2 Painting. The parts and assemblies shall be painted as specified on the drawing and in accordance with MIL-STD-171. The visible surfaces of attaching hardware (nuts, screws, washers, etc.), exposed unpainted surfaces of mounting faces, or blemished areas shall be primed and painted to match surrounding surfaces after final assembly. To avoid necessity of painting exposed surfaces of cadmium plated hardware (including screws, nuts and washers), the hardware may be dyed with a supplementary chromate finish, Forrest Green, equivalent to color number 34079 of Federal Standard No. 595.

3.2.5 Welding. Interpretation of weld symbols specified on the applicable drawings shall be as defined in AWS A2.4. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens of the applicable welding specifications. Qualification of arc welding procedures and welders shall be in accordance with MIL-STD-248. Qualification of inertia friction welding shall be in accordance with MIL-STD-1252. All plug welds shall be flush and free of snags or protrusions. Grinding flush is not required unless specified on the drawing or in the welding specification. Weld splatter and similar excess weld shall be removed.

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3.2.5.1 Arc welding, aluminum alloy. Welding of aluminum alloy parts shall conform to MIL-W-45205. Type 5356 welding electrode of MIL-E-16053 shall be used for 5000 series aluminum alloy welds. Type 5183 or 5556 welding electrodes of MIL-E-16053 may be used if Type 5356 welding electrode is unavailable. When welding aluminum alloy parts other than 5000 series aluminum alloy, the welding shall be in accordance with the specification referenced on the applicable drawings.

3.2.5.2 Arc welding, ferrous metals. Welding of steel parts and assemblies shall be in accordance with weld classification of MIL-STD-1261 referenced on the applicable drawings.

3.2.5.3 Welding, inertia. Inertia friction welding shall be in accordance with MIL-STD-1252.

3.2.5.4 Magnetic particle inspection. When magnetic particle inspection is indicated on the drawing, it shall be in accordance with MIL-I-6868.

3.2.5.5 Penetrant inspection. When penetrant inspection is indicated on the drawing, it shall be in accordance with MIL-I-6866.

3.2.5.6 Radiographic inspection. Radiographic inspection shall be in accordance with MIL-STD-00453 (USAF). Aluminum inspection shall be to the standards of Appendix A of this specification and steel inspection to the standards of Appendix B to this specification as referenced on the applicable drawings.

3.2.6 Soldering. Soldering of parts shall be as indicated on the applicable drawing and in accordance with MIL-S-6872.

3.2.7 Threads. Threads shall conform to the thread series and class indicated on the applicable drawing, and be in accordance with Handbook H28. Unless otherwise specified, coated or plated class 3 threads shall conform to the coated thread data specified in MIL-S-7742.

3.2.7.1 Torque, threaded fasteners. Unless otherwise specified on the applicable drawing, threaded fasteners shall be torqued in accordance with the following torque requirements:

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Thread Size	Torque
No. 4 - 48UNF	6-8 inch-pounds
No. 6 - 32UNC	7-9 inch-pounds
No. 8 - 32UNC	14-17 inch-pounds
No.10 - 24UNC	20-25 inch-pounds
No.10 - 32UNF	20-25 inch-pounds
.250 - 20UNC	70-80 inch-pounds
.375 - 16UNC	20-24 foot-pounds
.500 - 13UNC	50-60 foot-pounds
.625 - 11UNC	90-119 foot-pounds

3.3 Performance characteristics. The completed carriage shall conform to the characteristics specified on Drawing K12007600 and be in accordance with this specification. The carriage, with slave unit properly installed (see 6.2), shall meet the requirements specified in 3.3.1 through 3.3.8 when placed on a level surface. Testing shall be as specified in 4.5.3.2.

3.3.1 Traversing mechanism.

3.3.1.1 With the carriage placed in firing position, with the trails in open position and slave unit installed, the traversing mechanism shall be capable of traversing the top carriage 22-1/2 degrees + 1/2 degree left and right of center. The traverse stops shall not be removed during this test. The torque required to traverse the top carriage left and right of center shall not exceed 120 inch-pounds at the handwheel shaft.

3.3.1.2 The maximum backlash of the traversing handwheel shall not exceed 30 degrees of rotation in either direction of the handwheel while traversing a maximum of 22-1/2 degrees + 1/2 degree left and right of center.

3.3.1.3 With the trails in the closed position, the traversing mechanism shall be capable of traversing the top carriage 360 degrees in either direction.

3.3.2 Elevating mechanism. The elevating mechanism shall elevate the cannon tube of the slave unit from a position of minus 75 mils depression to plus 1275 mils in elevation. The torque required to elevate or depress at any point throughout the operating range shall not exceed 180 inch-pounds at the handwheel shaft. The maximum total system backlash of the elevating handwheel shall not exceed 30 degrees of rotation at any position in either direction of elevation or depression. Total system backlash is defined as the total handwheel lost motion until first movement of the cradle, when the elevating handwheels are utilized to elevate and depress the weapon.

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3.3.2.1 Elevating mechanism timing. The elevating mechanism shall be properly timed as evidenced by the right hand spline displaying a non-binding movement at the travel lock postion when tested as specified in paragraph 4.6.2.1.

3.3.3 Speed shift assembly. With the carriage and slave unit placed in firing position with trails in open position, the speed shift assembly shall be capable of raising the carriage and slave unit to permit manual rotation of the carriage 360 degrees. The maximum force required on the hydraulic pump handle to actuate the speed shift shall not exceed 19.0 pounds for the up position and 17.0 pounds for the down position. The speed shift assembly shall be capable of maintaining a controlled rate of descent to the firing base of not less than 5 seconds and not more than 20 seconds during gravity drop.

3.3.4 Wheel actuator. The hydraulic wheel actuator shall be capable of lowering the carriage to the firing position and raising the carriage to the travel position. The maximum force required on the hydraulic pump handle to raise and lower the carriage shall not exceed 32 pounds. During gravity drop of the carriage, a controlled rate of descent to the firing base shall not be less than 5 seconds nor more than 20 seconds.

3.3.5 Handbrakes. The handbrakes, when engaged, shall prevent wheel rotation while the carriage, with slave unit installed, is in an ascending or descending travel position on a 10 degree slope having a hard surface on which the tires will not slip. Travel position shall be defined as with the trails locked in closed position, and the lunette raised to a height of 30 inches above ground level. Upon releasing the handbrakes, the carriage shall be free to roll.

3.3.6 Gun elevation - plumb travel. The elevating trunnions of the carriage shall be cross leveled by having the slave unit cannon track a plumb line within 0.1 mil through a range of 0 to 978 mils elevation minimum 1275 mils.

3.3.7 Locking devices. All locking and fastening devices for securing component parts shall be positive in action.

3.3.8 Variable recoil mechanism. The variable recoil mechanism shall be assembled with the carriage and slave unit (see 6.2) and adjusted in accordance with Drawing K12007600 and K12008000. When elevating or depressing the cannon, the mechanism shall move freely without binding or looseness. Adjustment of the linkage shall be as follows:

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3.3.8.1 Short recoil. The 55 degrees (from vertical) mark on end of recuperator cylinder shall lie within the notch on end of regulator lever when the cannon is at 1025 mils elevation or less.

3.3.8.2 Long recoil. The 105 degrees (from vertical) mark on end of recuperator cylinder shall lie within the notch on end of regulator lever when the cannon is at 50 mils elevation or less.

3.4 Proof firing. The carriage, assembled with one M45 recoil mechanism (MIL-M-45986) and one M199 cannon (MIL-C-45975), shall withstand proof-firing as specified in Table I with no evidence of any system malfunction. Upon completion of proof-firing, there shall be no evidence of structural damage, permanent deformation or degradation of operating characteristics. Testing shall be as specified in 4.5.3.3.

TABLE I. Proof firing[1]

No. of Rounds	Prop Charge	Zone	Proj. Inert	Elevation (Mils) ± 10 Mils	Traverse	Recoil Length (inches)	Cycle Time (Sec.)
1	M4A2	7	2/	800	Max. Right	40-44	2 to 4
1	M203	8	2/	900	Max. Right	50-54	2 to 4
1	M203	8	2/	300	Max. Right	64-68	2 to 4
1	M4A2	7	2/	800	Max. Left	40-44	2 to 4
1	M203	8	2/	900	Max. Left	50-54	2 to 4
1	M203	8	2/	300	Max. Left	64-68	2 to 4

NOTES:
1/ All charges shall be conditioned and fired at 135°F to 145°.
2/ Any standard projectile loaded to 103 ± 1 pounds.

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3.5 Road test. The carriage, assembled with the M45 recoil mechanism (MIL-M-45986) and M199 cannon (MIL-C-45975) shall be capable of withstanding the following 50 mile road test without damage, permanent deformation or loosening of parts. Testing shall be as specified in 4.5.3.4.

a. 35 miles at speeds between 30 and 55 mph over hard surface or improved roads.

b. 10 miles between 10 and 15 mph over secondary roads.

c. 5 miles between 5 and 10 mph over unimproved or cross-country roads.

3.5.1 Brake system. The brake system, when the carriage with the slave unit installed, is attached to the prime mover, shall provide effective and controlled braking of the Howitzer when stopping or slowing down from speeds up to 55 mph on improved roads and when descending slopes up to 60 percent. The break away emergency action of the brakes shall be capable of locking the wheels when the Howitzer and air hoses break away from the prime mover.

3.6 Reliability, maintainability and durability. Carriages manufactured in accordance with this specification shall be free from any defects that compromise, limit or reduce the end item howitzer or component capability specified herein and shall be subject to examination and test by the Government to prove howitzer capabilities and performance in accordance with the following:

3.6.1 Reliability. The reliability requirement is a specified value of 1100 mean rounds between failure and a minimum acceptable value of 700 mean rounds between failure as defined in AR 702-3.

3.6.2 Maintainability. The maintainability requirement is 0.5 hours for mean time to repair (MTTR) at organizational and 2.0 hours for direct support.

3.6.3 Durability. The durability requirement is 15,000 Equivalent Full Charge (EFC) rounds without rebuild or overhaul.

3.7 Identification and instructional markings. Identification and instructional markings shall be applied in accordance with Drawing K12007600. Other marking, including application of part numbers and instructional marking, shall be as specified on the applicable drawings and in accordance with MIL-STD-130. Serial numbers shall be furnished by the procuring activity (see 6.1).

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3.8 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations and missing operations which may affect serviceability, functioning, operation, appearance or safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, quality assurance provisions, MIL-W-63150 and such documents as specified in LQ12007600 and the contract. These minimum inspections shall not be costrued as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the contract, drawings, and specifications.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)), shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to

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the quality conformance inspection specified herein and such other inspection as necessary to determine that all requirements of the contract have been complied with. One of the first article items will be subjected to a reliability, maintainability and durability demonstration (see 4.7), performed by the Government for informational purposes.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size, and presentation of inspection lots shall be in accordance with MIL-STD-105. Carriages shall be assembled from lots of component parts that have complied with all inspection requirements specified herein.

4.5.2 Examination.

4.5.2.1 Carriages. Visually and manually examine each carriage to determine conformance with the applicable requirements. Each step in the examination shall include a visual examination for proper cleaning, the presence of the specified protective coating, and to determine the general quality, completeness of manufacture, assembly and workmanship (see 3.8). Visually examine all markings to assure that they are correct and legible (see 3.7).

4.5.2.2 Parts and assemblies. Parts and assemblies shall be visually and manually examined to determine compliance with the materials, construction and design requirements of 3.2 and the applicable requirements of 3.2.1 through 3.2.7. Visually examine for marking (see 3.7) and workmanship (see 3.8). Parts and assemblies failing to comply with the requirements shall be rejected.

4.5.2.2.1 Heat treatment. Examination and tests of heat treated components shall be in accordance with MIL-H-6088 or MIL-H-6875, as applicable.

4.5.2.2.2 Surface texture. When surface texture is specified on the drawing, examination and interpretation shall be in accordance with ANSI B46.1.

4.5.2.2.3 Welds and welding. Examination and tests of welded parts and welding procedures for aluminum alloys shall be in accordance with MIL-W-45205. Examination and tests of welding procedures for steel parts shall be in accordance with MIL-STD-248 or as otherwise specified on the applicable drawings.

4.5.2.2.4 Magnetic particle inspection. When magnetic particle inspection is specified on the drawing, the inspection shall be in accordance with MIL-I-6868.

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4.5.2.2.5 Penetrant inspection. When penetrant inspection is specified on the drawing, the inspection shall be in accordance with MIL-I-6866.

4.5.2.2.6 Radiographic inspection. When radiographic inspection is specified on the drawing, the inspection shall be in accordance with the specifications referenced on the drawing.

4.5.2.2.7 Protective finishes. Examination of painted surfaces and inorganic finishes shall be in accordance with the specifications specified on the applicable drawing and MIL-STD-171.

4.5.2.2.8 Soldering. Examination and test of soldered parts shall be as specified in MIL-S-6872.

4.5.2.2.9 Marking. Visually examine the identification plate for compliance with the applicable drawing, for secure assembly, serialization and to assure that all pertinent data has been applied.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete carriage. If test requirements cited herein are not met, acceptance of the carriage or carriage lot shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.

b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

c. Examine carriages, partially assembled carriages and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned carriages for retest.

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4.5.3.2 Performance characteristics testing. Each completely assembled carriage shall be tested for performance characteristics (see 3.3 and 3.3.1 through 3.3.8) using the test methods specified in 4.6. Failure of the carriage to comply with the requirements shall be cause for rejection.

4.5.3.3 Proof firing testing. Unless otherwise specified (see 6.1), one carriage assembly shall be selected from each carriage lot and shall be subjected to proof-firing in accordance with 3.4 using the test method specified in 4.6.9. Failure of the tested carriage to comply with the requirements shall be cause for rejection of the lot. After correction of problem areas as specified in 4.5.3.1, the resubmitted lot shall be tested as specified above except that the sample size shall be three.

4.5.3.4 Road testing. Each carriage which is proof-fired in accordance with 4.5.3.3 shall also be subjected to a road test and brake system test to determine compliance with 3.5 and 3.5.1 using the test method specified in 4.6.10. Failure of the carriage to comply with the requirements shall be cause for rejection of the lot. Retesting shall be as specified in 4.5.3.3.

4.5.4 Inspection equipment. Unless otherwise specified herein, responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on List of Inspection Equipment Numbers EL12007600-I, and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5 Inspection of packaging. Unless otherwise specified (see 6.1), inspection to determine compliance with preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

4.6 Test methods.

4.6.1 Traversing mechanism test. A torque wrench graduated in inch-pounds shall be utilized for measuring loads at the shaft of the handwheel. With the slave unit installed, the carriage in enplaced position and level, and the slave unit cannon tube at zero degrees elevation, traversing measurements shall be taken at 0°, $\pm 15^{o}$ and $\pm 22^{o}$ while traversing the carriage in both directions.

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4.6.1.1 Traversing mechanism backlash measurement test. The following procedure shall be utilized to measure left and right system backlash:

A. Equipment utilized:

1) A circular (O.D. same diameter as traverse handwheel) template with clearance cut-outs for the handle, located 180 degrees from each other, with degree markings on the front face, showing a span of 50 degrees and graduated in one degree intervals located left and right of the zero mark and two mounting clamps on the rear face.

2) A pointer with an attached spring and chain.

3) A dial indicator (with a range per revolution of .100 inch and a travel range of 1.00 inch) attached to a magnetic base with an adjustable arm extension.

4) The large circular template utilized for the elevation backlash test.

B. Assembly of the equipment to the Carriage to be tested:

1) The circular template is centrally mounted over the traverse handwheel and secured with the clamp provided.

2) The pointer is mounted behind the traverse handwheel and secured in place by pulling the chain firmly around the handwheel housing and hooking the chain to the notch provided.

3) The large circular template that is utilized for the elevation backlash test is mounted on the left hand elevation handwheel.

4) The dial indicator, mounted on the magnetic base with adjustable arm/extension, is placed on the left hand top surface of the Bottom Carriage, adjacent to the traverse stop, with the indicator point contacting the large circular template on the elevation handwheel at right angles.

C. Backlash inspection procedure:

1) With the carriage placed in firing position with trails in open position and slave unit installed, gun tube level, and the "V" notch on the rear of the top carriage base aligned at 400 mils on the azimuth plate located at the rear of the bottom carriage, turn the traverse handwheel clockwise to align the zero degree mark on the circular template with the pointer.

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2) Holding the handwheel steady, position the magnetic base to depress the dial indicator point (use at least 3 revolutions of travel) and orient the indicator point at right angles to the circular template on the elevation handwheel. Set the indicator dial at zero.

3) While watching the dial, turn the handwheel clockwise at least one-and-a-third turns (on the dial).

4) While watching the dial indicator, turn the handwheel in a counterclockwise direction and stop when the dial indicator pointer returns the one-and-a-third revolutions (on the dial) back to the original zero reading. Read the number of degrees of backlash on the circular template aligned with the pointer.

5) Record the reading on a suitable inspection form.

6) Remove the magnetic base and dial indicator from the bottom carriage.

7) Repeat steps one through six at 670 and 780 mils on the azimuth plate.

8) Traverse the top carriage to 20 mils on the azimuth scale.

9) Repeat steps one through six at 20 mils and 130 mils on the azimuth scale.

4.6.2 Elevating Mechanism Test. A torque wrench graduated in inch-pounds shall be utilized for measuring loads at the shaft of the left handwheel. Measurements shall be taken at the following elevations, both in elevating and depressing: -4.2^{o} (-75 mils), $+5^{o}$ (89 mils), $+10^{o}$ (178 mils), $+15^{o}$ (267 mils), $+25^{o}$ (444 mils), $+35^{o}$ (622 mils), $+45^{o}$ (800 mils), $+55^{o}$ (978 mils), $+65^{o}$ (1156 mils) and $+70^{o}$ (1245 mils). The method for taking measurements is as follows: With the slave unit installed and the carriage in emplaced position and level, depress to -75 mils and take the torque reading on the nut clockwise or counterclockwise. Then elevate to the next station and repeat taking torque readings on the nut clockwise or counterclockwise. Repeat this operation at all stations while elevating the cradle and slave unit between -75 mils and maximum elevation +1245 mils.

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4.6.2.1 Elevating Mechanism Backlash Measurement Test. The following procedure shall be utilized to measure left and right system backlash:

A. Equipment utilized:

1) A circular template, approximately 15 inches in diameter, with clearance cut-outs for the handle, located 180 degrees from each other, with degree markings on the front face, showing a span of 50 degrees and graduated in one degree intervals, located left and right of the zero mark and two mounting clamps on the rear face.

2) A pointer with an attached spring and chain.

3) A dial indicator (with a range per revolution of .100 inch and a total range of 1.00 inch).

4) A strap clamp with a quick release tension hook.

5) A split clamp with a quick release tension hook and with a contact surface for the dial indicator point.

B. Assembly of test equipment to the carriage to be inspected:

1) The circular template is mounted over the left elevating mechanism handwheel by locating the shoulder, on back of the template, into the nut counterbore and securing to the handwheel with both mounting clamps on the rear face of the template.

2) The pointer, for the circular template, with an attached spring and chain, is mounted behind the same handwheel and secured to the handwheel housing by pulling the chain firmly around the handwheel housing and hooking the chain in the notch provided as part of the pointer assembly.

3) The dial indicator is mounted to the strap clamp and then mounted on the lower end of the left cylinder (ref. dwg. 12007737) of the left elevating screw assembly (ref. dwg. 12007700).

4) The split clamp, with a quick release tension hook, is clamped in place on the lower end of the left tube (ref. dwg. 12007745) of the same elevating screw assembly just below the dial indicator with the contact surface of the clamp aligned with and against the dial indicator point. NOTE: This clamp is relocated on the tube after each elevation or depression setting of the cannon tube.

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C. Backlash Inspection Procedure:

1) Ensure that the flat on top of the breech block is clean.

2) Set the gunner's quadrant to the desired elevation setting and place it on top of the breech block flat with the LINE OF FIRE arrow pointed toward the muzzle.

3) Disengage the clutch by pressing down on the clutch handle. Elevate the weapon by turning the elevating handwheel, on either side of the carriage, until the gunner's quadrant bubble centers to the first elevation. Rotate gunner's quadrant end for end. The gunner's quadrant bubble should recenter.

4) Turn the handwheel counterclockwise until the pointer is aligned with the zero degree marking on the circular template. While holding the handwheel steady at zero degrees, set the dial indicator to zero.

5) While observing the dial indicator, continue turning the handwheel counterclockwise and stop when the dial indicator pointer has made at least one-and-a-third revolutions around the dial past the original zero setting (approximately .130 inch).

6) Continue watching the dial indicator and turn the handwheel in a clockwise direction and stop when the dial indicator pointer returns the one-and-a-third revolutions back to the original zero reading. Read and record the number of degrees of backlash on the circular template aligned with the pointer.

7) Read and record backlash readings at the following elevations and depressions: -75 mils, +89 mils, +267 mils, +622 mils, +978 mils and +1245 mils.

8) Repeat the above procedure for the right side, reassembling the test equipment on the right side, for each elevation and depression as specified in step 7, in mils, taking care to move the split clamp, when required, to preclude damage to the dial indicator.

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4.6.2.1.1 Elevating mechanism timing test. The testing method for timed elevating screw assemblies of the elevating mechanism shall be performed, with the slave unit installed, the cradle at the travel lock position, a balanced equilibrator system and the right hand spline exposed. The test shall be performed as follows:

Remove the travel lock pins and the traverse pin. Disengage the clutch and elevate/depress the gun tube above and below the travel lock position for ease of movement. Return cradle to travel lock position and secure travel lock pins which shall enter freely. Disengage clutch for approximately 2 to 8 seconds and then re-engage clutch. Check right hand shaft utilizing steady, even handed pressure for the requirement as specified in paragraph 3.3.2.1.

4.6.3 Speed shift assembly test. The speed shift assembly is actuated by moving the hydraulic selector valve to the "DOWN" position. With the slave unit installed and the carriage in enplaced position, elevate or depress the cannon tube to approximately 800 mils elevation to provide a balance for speed shifting the carriage and slave unit. Pump one or both hydraulic pumps to lower the speed shift, permitting the carriage and slave unit to be raised off the ground. Lift the right and left trails and pivot the carriage and slave unit 360°. Move the selector valve to the "UP" position. A controlled rate of descent to the firing base of not less than 5 seconds and not more than 20 seconds shall be maintained during the gravity drop phase. Pump the hydraulic pumps to retract the speed shift to allow the carriage and slave unit to be placed on the ground.

4.6.4 Wheel actuator test. The completed carriage shall be assembled with the slave unit in the towed position (travel lock engaged) and the parking brakes disengaged. Levers for the speed shift and wheels shall be in the "OFF" position. Turn lever to "WHEELS DOWN" and pump to release weight from the lock pins. Release the lock pins. Turn lever to "WHEELS UP" position. Carriage shall settle to base plate in not less than 5 seconds and not more than 20 seconds. Pump until the wheels go up to the firing position and engage lock pins. Turn wheel lever to "OFF"; turn the lever to "WHEELS UP". Pump to release weight from the lock pins. Release the lock pins. Turn the lever to "WHEELS DOWN" and pump until the wheels are in travel position and engage in lock pins.

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4.6.5 Handbrakes test. The handbrakes of the completed carriage with the slave unit installed shall be tested to determine compliance with 3.3.5. The requirement may also be checked by applying 2650 + 50 pound push or pull on the lunette with the carriage and slave unit in traveling position on a level surface.

4.6.6 Gun elevation - plumb travel test.

4.6.6.1 Test equipment and instruments. The following test equipment and instruments shall be used:

a. Mechanical jacks with a minimum of one square foot base for leveling the carriage trunnions.

b. A plumb line, approximatley 1/16 inch in diameter and about 45 feet long, shall be suspended one foot or less from the slave unit cannon tube. The line shall be shielded from wind currents and the plumb bob shall be immersed in a container of oil to act as a dampener.

c. The borescope used for cannon tube elevation plumb travel shall have a cross hair reticle mounted in a self-aligning adapter to be inserted into the breech of the cannon so that the reticle has a focusing range of 5 meters to infinity. The reticle of the telescope shall be collimated with respect to the outside diameter of the adapter within 0.1 mil. The borescope shall have magnification of 4 power or higher.

4.6.6.2 Test procedure. Each completed carriage, with the slave unit installed, shall be tested through the elevation range specified in 3.3.6. With the slave unit installed, the carriage trunnion shall be cross-leveled. Using the borescope, the cannon tube shall track a plumb line throughout the elevation range and deflection limit specified in 3.3.6.

4.6.7 Locking devices test. All Basic Issue Items (BII) which attach to the carriage shall be installed and tested for fit of brackets and operation of all locking and fastening devices.

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4.6.8 Brake system test. The carriage, with the slave unit installed, shall be towed at a speed of not less than five miles per hour over a hard, smooth surface. Application of the brakes shall cause the wheels to lock and slide. The break away emergency action of the brakes shall be checked by disconnecting the air hose couplings with the drain cock closed. The brakes shall apply immediately, causing the wheels to slide.

4.6.9 Proof firing test. Each carriage with M45 Recoil Mechanism and M199 Canon selected as specified in 4.5.3.3 shall be proof-fired. Visual examination for cracks, weld failure, deformations and other forms of damage shall be conducted on each carriage at the conclusion of the proof-firing test.

4.6.10 Road test. After proof firing, the carriage (with Recoil Mechanism and M199 Cannon) in traveling position with telescopes (if applicable) in stowed position (less dummy round) shall be coupled to a vehicle equipped with standard pintle and road tested at the speeds and road conditions specified in paragraph 3.5.1. Application of the brakes shall cause the wheels to lock and slide. The break away emergency action in the brakes shall be tested by disconnecting the air hose couplings with the drain cock closed. The brakes shall apply immediately, causing the wheels to slide. Visual examinations for cracks, weld failures, deformations and other forms of damage shall be conducted.

4.6.11 Variable recoil mechanism test. Check the installation and adjustment of the variable recoil linkage. Recuperator cylinder marks at 105° and 55° shall remain visible within the notch on the regulator level during repeated cycling between minimum and maximum cannon tube elevations. The variable recoil mechanism shall move freely without binding or excessive looseness throughout the entire range of travel.

4.7 Reliability, maintainability and durability demonstration. The howitzer demonstration will be conducted in accordance with the operational mode summary and reliability failure definition in the M198 scoring criteria. The test will be conducted by the Government for informational purposes and shall provide the equivalent of a 2,500 EFC round reliability test and a 3,000 EFC round durability test. The two tests may be combined. Maintenance data will be collected according to the scoring criteria, and the MTTR will be calculated according to the scoring criteria. Howitzers exhibiting a reliability, maintainability or durability failure shall be cause for the contractor to perform a failure analysis in accordance with 4.5.3.1. The Government representative shall verify the results of the failure analysis and assure deficiencies causing the failure have been corrected on all weapons and components in process at the contractor's plant.

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5. PACKAGING

5.1 Pilot pack. The pilot pack shall consist of a complete carriage preserved in accordance with the requirements specified herein for the level of protection specified (see 6.1), packed Level B and forwarded as specified in 3.1.

5.2 Preservation. Preservation shall be Level A or B, as specified (see 6.1). Materials, methods, processes and procedures specfied herein shall conform to the requirements of MIL-P-116 and MIL-P-14232. Materials, of a minimum size in accordance with MIL-P-14232, shall conform to the requirements of the following specifications to the extent specified herein:

a. Barrier material: Type I, Grade A, Class 2 of MIL-B-121.

b. Overwrap: Type I of MIL-P-130.

c. Osnaburg cloth: Class 2 of CCC-C-429.

d. Strippable compound: Type II, Class 1 of MIL-C-16555.

e. Paperboard: Style 2, Type III of PPP-P-291.

f. Plywood: Group B (PS-1) std. int. w/ext. glue of NN-P-530.

g. Strapping: Class 2, Finish B, Grade 2, 16 Gauge of QQ-S-781.

h. Tapes: PPP-T-42 and Type III, Class 2 of PPP-T-60.

5.2.1 Level A.

5.2.1.1 Cleaning. Unless otherwise specified herein, all unpainted ferrous surfaces shall be cleaned by process C-3. Painted and non-ferrous surfaces shall be cleaned by process C-1. Hydraulic system assemblies and components shall be cleaned in accordance with the applicable drawing requirements.

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5.2.1.2 Drying. Unless otherwise specified herein, all cleaned surfaces shall be dried by any applicable procedure of MIL-P-116 which is non-injurious to the item.

5.2.1.3 Preservative application. Oils or lubricants removed or contaminated during cleaning shall be reapplied in accordance with applicable drawing requirements. Unpreserved exterior metallic surfaces shall be preserved with grease, Type P-11 or MIL-P-116. All interior access openings of hydraulic system assemblies and components shall be sealed with caps/plugs conforming to MIL-C-5501. Strippable compound conforming to MIL-C-16555, when specified herein, shall be sprayed on osnaburg cloth conforming to CCC-C-429 in successive layers until a thickness of 0.035 ± 0.005 inch is obtained. The compound shall overlap all edges onto adjacent surfaces at least one inch.

5.2.1.4 Unit pack. The unit pack for each carriage shall be Method 1 and as specified herein.

5.2.1.4.1 Mount assembly.

5.2.1.4.1.1 Cradle assembly.

a. Preserve exposed, unpainted metallic surfaces with grease, Type P-11. Wrap preserved surfaces with barrier material conforming to MIL-B-121. Secure wrap with tape conforming to PPP-T-42.

b. Overwrap barrier material with cloth conforming to CCC-C-429. Secure overwrap with tape conforming to PPP-T-60.

c. Overspray the cloth with strippable compound conforming to MIL-C-16555.

d. Cut a slit approximately one inch long on the underside of the wrapped and sprayed areas to allow for drainage of entrapped moisture.

5.2.1.4.1.2 Variable recoil assembly.

a. Preserve exposed, unpainted metallic surfaces with grease, Type P-11. Wrap preserved surfaces with barrier material conforming to MIL-B-121. Secure wrap with tape conforming to PPP-T-42.

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b. Overwrap barrier material with cloth conforming to CCC-C-429. Secure overwrap with tape conforming to PPP-T-60.

c. Overspray the cloth with strippable compound conforming to MIL-C-16555.

d. After the strippable compound is thoroughly dry, secure the preserved variable recoil assembly to the cradle with tape conforming to PPP-T-60.

5.2.1.4.1.3 Equilibrators.

a. Preserve exposed, unpainted metallic surfaces with grease, Type P-11. Wrap preserved surfacers with barrier material conforming to MIL-B-121. Secure wrap with tape conforming to PPP-T-42.

b. Overwrap barrier material with cloth conforming to CCC-C-429. Secure overwrap with tape conforming to PPP-T-60.

c. Overspray the cloth with strippable compound conforming to MIL-C-16555.

d. Wrap the bellows portions with paperboard conforming to PPP-P-291. Secure wrap with tape conforming to PPP-T-60.

5.2.1.4.2 Top carriage.

a. Preserve exposed, unpainted metallic surfaces with grease, Type P-11. Wrap preserved surfaces with barrier material conforming to MIL-B-121. Secure wrap with tape conforming to PPP-T-60.

b. Elevating and traversing mechanism. Preserve exposed, unpainted metallic surfaces of shafts, couplings and universal joints with grease, Type P-11. Wrap preserved surfaces with barrier material conforming to MIL-B-121. Secure wrap with tape conforming to PPP-T-60.

5.2.1.4.3 Bottom carriage. Preserve exposed, unpainted metallic surfaces with grease, Type P-11. Wrap preserved surfaces with barrier material conforming to MIL-B-121. Secure wrap with tape conforming to PPP-T-60.

5.2.2 Level B.

5.2.2.1 Cleaning, drying and preservation application. Cleaning, drying and preservative application shall be the same as specified for Level A (see 5.2.1.1, 5.2.1.2 and 5.2.1.3).

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5.2.2.2 Unit pack. The unit pack for each carriage shall be the same as specified for Level A (see 5.2.1.4), with the following exception: Delete the overwrap of osnaburg cloth (CCC-C-429) and strippable compound (MIL-C-16555) and substitute an overwrap conforming to MIL-P-130, secured to adjacent surfaces with tape conforming to PPP-T-60.

5.3 Packing.

5.3.1 Levels A and B. The carriage shall be prepared in tow-away position, as specified herein, including the following:

a. Trail lock handle: Secure the trail lock handle in closed position with strapping conforming to QQ-S-781.

b. Tires: Inflate carriage tires to 10 pounds above normal operating pressure for shipping and stowage purposes.

c. Strap assemblies: Compactly fold cloth equipment straps and secure to adjacent surfaces with tape conforming to PPP-T-60.

d. Locking pins: Secure all locking pins in place with tape conforming to PPP-T-60 and cap ball ends with specified tape.

5.4 Marking.

5.4.1 Levels A and B. Identification marking for the carriage shall be applied in accordance with the requirements of MIL-STD-129 on a plywood panel conforming to NN-P-530, size 3/8 x 14 x 20 inches. The panel shall be secured to the carriage with strapping conforming to QQ-S-781.

6. NOTES

6.1 Ordering data. Procurement document should specify the following:

a. Title, number and date of this specification.

b. List of drawings, specifications and publications pertinent to the carriage, showing applicable revision dates.

c. Shipping instructions for first article and pilot pack (see 3.1, 4.4. and 5.1).

d. Block of serial numbers (see 3.7).

e. Packaging inspection, if different (see 4.5.5).

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f. Selection of applicable level of protection, A or B (see 5.1, 5.2 and 5.3).

g. Number of carriages required for proof firing, if other than as specified (see 4.5.3.3).

6.2 Slave unit. The slave unit consists of one M45 recoil mechanism (MIL-M-45986) and one M199 cannon (MIL-C-45975), or an acceptable equivalent.

6.3 Radiographic standards.

ATAC-STD-113	-Reference standards and radiographic procedures for partial penetration aluminum welds. The reference standards were prepared by the U.S. Army Tank Automotive Command under project authorization of the U.S. Army Materials and Mechanics Research Center.
ATAC-STD-114	-Reference standards and radiographic procedures for partial penetration steel welds. The reference standards were prepared by the U.S. Army Tank Automotive Command under project authorization of the U.S. Army Materials and Mechanics Research Center.

6.4 Supersession data. This specification supersedes MIL-C-45987 (WC), dated 15 February 1977, with Amendment 5, dated 07 October 1982, Carriage, 155MM Howitzer: M39.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1025-A140)

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10. SCOPE

10.1 This procedure provides standard reference radiographs for radiographic inspection of partial penetration weldments in aluminum plate extrusions, castings or forgings up to two (2) inches thick. They also apply to full penetration welds as, applicable.

10.2 Classification. The reference radiographs comprised in this standard are intended primarily for use on welds made with the inert gas metal-arc process.

20. APPLICABLE DOCUMENTS

SPECIFICATIONS

MILITARY

MIL-W-45205A(MR)	-Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.

STANDARDS

MIL-STD-410	-Nondestructive Testing Personnel, Qualification and Certification
MIL-STD-00453 (USAF)	-Inspection, radiographic.
ATAC-STD-113	-Reference standards and radiographic procedures for partial penetration aluminum welds. (Only to the extent specified herein.) Prepared by the U.S. Army Tank Automotive Command.
ATAC-STD-114	-Reference standards and radiographic procedures for partial penetration steel welds (Only to the extent specified herein.) Prepared by the U.S. Army Tank Automotive Command.

30. DEFINITIONS

30.1 Partial Penetration Joint Design. The term partial penetration as used in this standard is defined as a weld joint containing an intentionally unfused area in the central portion of the joint.

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40. GENERAL REQUIREMENTS

40.1 Radiographic Inspection. Weldments shall be radiographically inspected in the locations specified on the applicable radiographic position chart drawings, component drawings, specification, or as specified in the contract. Laboratories and personnel performing radiographic inspection shall meet the requirements of MIL-STD-00453 and MIL-STD-410, as applicable.

40.1.1 Radiographic Frequency. All aluminum weldments where the R.T. symbol is shown on the drawings shall be radiographed 100% unless drawing notes associated with the radiographic examination states otherwise.

40.2 Radiographic Variance. Because specialized techniques are required to facilitate radiographic inspection, it should be noted that some discontinuites may be masked due to the configuration of the specific joint design. For this reason, the joints will be inspected as follows:

40.3 Weld Designation: Inspection Categories.

1. Category 1 -Radiograph to be inspected for scatter porosity, cluster porosity, linear porosity, gas cavities, incomplete fusion, and cracks.

2. Category 2 -Radiograph to be inspected for scatter porosity, cluster porosity, linear porosity, gas cavities, incomplete fusion, inadequate penetration and cracks.

40.4 Inspection Acceptance Standard. "Reference Radiographs ATAC-STD-113, ATAC-STD-114 apply" to the extent specified below.

a. Scatter porosity, (fine and coarse): STD 2-AL-49L-2C
b. Linear porosity: STD 1-AL-61L-1
c. Cluster porosity: STD 1-AL-3-1
d. Gas Cavities: Shall not exceed 1/8" dia. with a pore frequency no greater than STD-1-AL-45-1
e. Inadequate penetration and incomplete fusion:

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APPENDIX A

RS 12007600 AL

1. Weld and Parent Joint Material cross sections (radiation path) less than 1/2":

(a) 1/32" width or less rule - Indications in the root area shall be acceptable if the indications are less than or equal to a 1/32 inch in width, regardless of length in the root area.

(b) Greater than 1/32" but not greater than 1/16" in width, and length rule - Indications in the root area shall be acceptable if the indications are greater than 1/32 inch in width but not greater than 1/16 inch in width and the total length of the indications are not greater than 1T for any 12T length of weld (T is the thickness of the thinnest parent metal member of the weld joint.

(c) Greater than 1/16" width rule - Indications in the root area are not acceptable that are greater than 1/16 inches in width, regardless of length in the root area

2. Weld and Parent Joint Material cross section (Radiation path) 1/2" and greater:

(a) Thirty percent (30%) size and length rule - Indications shall be acceptable in the root area if they have a width not in excess of 30% of the thinnest parent metal member of the weld joint, (but not to exceed 1/16 inch in width)

and

the total length of the indications in the root area are not greater than 1T for any 12T length of weld (T is the thickness of the thinnest parent metal member of the weld joint).

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(b) Greater than 1/16" width rule - Indications in the root area are not acceptable that are greater than 1/16 inches in width, regardless of length, in the root area.

SEE LAST PAGE OF APPENDIX A FOR EXAMPLES FOR THE INADEQUATE PENETRATION AND INCOMPLETE FUSION RULES.

f. Cracks are not acceptable.

g. Foreign inclusions: All foreign material which have become an integral part of the weld shall be judged as follows:

1. Rounded indications shall be judged as per STD-2-AL-49L-2C, STD-1-AL-61L-1 and STD-1-AL-3-1.

2. Linear indications shall be judged as per the slag inclusion standards of steel welds (ATAC-STD-114), STD-1-FE 48L-1.

40.4.1 Rounded indications of minor axis under 1/32 inch are not to be evaluated. Rounded indications have a major to minor axis dimensional ratio of three or less. They may be irregular in shape and may have tails which shall be included in evaluating the size.

40.4.2 Rejected welds shall be repaired and reinspected by radiography to the original standard.

40.4.3 Various Flaw Types. With the exception of cracks, the common internal flaws experienced in aluminum weldments made with the inert gas metal-arc process, are illustrated in the reference standards forming part of this procedure. All cracks are rejected.

a. Scattered porosity (fine and coarse). This flaw consists of scattered voids formed by gases failing to escape during weld metal solidification. On the radiograph, they are dark, round or elongated spots of varying size and density.

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b. Clustered porosity. Clustered porosity appears the same as scattered porosity, except the pores are concentrated in one area, generally at the start of a weld bead or an interrupted arc.

c. Inclusions. This flaw is derived from a weld bead contaminent such as steel wire bristle.

d. Linear porosity. The type most common in welds results from expanding gases generated in the normally unfused area. The cavities are linearily distributed at the root of the weld deposit and generally range from 1/32 to 1/16 inch diameter.

e. Gas cavities. This flaw generally results from inadequate shielding gas or severe contamination of the gas. (It appears the same as scattered porosity except the average cavity is approximately 1/8 inch diameter).

f. Inadequate penetration. In partial penetration and full penetration welds, this flaw consists of a linear root void in excess of the unfused area normally present. (It appears on the radiograph as a straight dark line at either or both roots.)

g. Incomplete fusion. Incomplete fusion is a failure of the weld metal to fuse completely with base metal or proceding bead. The first condition is the most common in aluminum welds and is generally associated with inadequate penetration near the root of the weld. (This flaw is very difficult to detect radiographically, due to its hairline condition and frequent interference from inadequate penetration indications).

40.5 Departures from the inspection categories (paragraph 4.3) and the acceptance standard (paragraph 4.4) shall be as noted for the specific radiographic position in the Radiographic Classification Tables of this radiographic standard.

50. DETAILED REQUIREMENTS

50.1 Application of reference standards.

50.1.1 Determination of acceptability. Acceptability of production welds shall be determined by directly comparing the production radiograph with the designated reference standard for each type of flaw. The extent of flaw exhibited in the designated standard will be permitted throughout the length of

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any particular weld joint provided the flaw content in any weld length equal to that of the reference standard does not exceed that shown in the standard. Exceptions to this are required for inadequate penetration and incomplete fusion because acceptability of these flaws is based on material thickness.

50.1.2 Evaluation of single flaws. When the production radiograph shows flaws which are equal to or less severe than the designated acceptance standard specified for those types of flaws, the weld shall be considered radiographically acceptable. If the radiograph shows flaws of greater severity than specified, the weld shall be rejected, unless repair is permitted.

50.1.3 Multiple flaws.

a. When the production radiograph shows one predominant type of flaw to an extent equal to the designated acceptance standard, and all other types to be less severe than the designated standard, the weld shall be considered radiographically acceptable.

b. When the production radiograph shows two types of flaws, to an extent equal to the designated acceptance standard for each type of flaw, the weld may be accepted, however, an investigation shall be immediately instituted to correct the welding process or processes that resulted in the multiple flaws being equal to the standards. Subsequent radiographs of this weldment on the next succeeding unit displaying two types of flaws equal to the standard shall be rejected. A frequency of no more than one weldment in a total of twenty identical weldments fabricated shall be considered acceptable under this criteria.

c. When the production radiograph shows three or more types of flaws to an extent equal to the designated acceptance standard for each type of flaw, the weld shall be rejected.

50.1.4 False indications. At times, radiographic films may indicate weld flaws when actually the film is defective. If doubt exists as to whether the flaw is in the weld or a film imperfection, the weld shall be radiographed again.

50.2 Inspection records. Radiographic reports and films shall be maintained and made available upon request to the designated government agency in accordance with MIL-STD-00453.

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APPENDIX A

RS 12007600 AL

RADIOGRAPHIC CLASSIFICATION FOR ALUMINUM WELDMENTS

Radiographic Position No.	Category	
7903-1A	1	See Notes 1 & 3
7903-1B	1	See Notes 1 & 3
7903-2A	1	See Notes 1 & 3
7903-2B	1	See Notes 1 & 3
7903-3A	2	See Note 3
7903-3B	2	See Note 3
7903-4A	2	See Note 3
7903-4B	2	See Note 3
7924-1	2	
7924-2	2	
8101-1A	1	See Notes 1 & 3
8101-1B	1	See Notes 1 & 3
8101-1C	1	See Notes 1 & 3
8101-1D	1	See Notes 1 & 3
8101-2A	1	See Notes 1 & 3
8101-2B	1	See Notes 1 & 3
8101-2C	1	See Notes 1 & 3
8101-2D	1	See Notes 1 & 3
8101-3A	1	See Notes 1 & 3 and Note 4
8101-3B	1	See Notes 1 & 3 and Note 4
8101-4A	1	See Notes 1 & 3 and Note 4
8101-4B	1	See Notes 1 & 3 and Note 4
8101-5A		Process Control
8101-5B		Process Control
8101-6A		Process Control
8101-6B		Process Control
8101-17A	1	See Notes 1 & 3 and 4
8101-17C	1	See Notes 1 & 3 and 4
8101-18A	1	See Notes 1 & 3 and 4
8600-1	2	See Note 2
8600-2	2	See Note 2
8600-3	2	See Note 2
8600-4	2	See Note 2
8601-1	2	See Note 2
8601-2	2	See Note 2
8601-3	2	See Note 2
8601-4	2	See Note 2
9107-1	2	See Note 2
9107-2	2	See Note 2
9107-3	2	See Note 2

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Radiographic Position No.	Category (Cont'd)
9107-4	2 See Note 2
9108-1	2 See Note 2
9108-2	2 See Note 2
9108-3	2 See Note 2
9108-4	2 See Note 2

NOTES:

Note 1. Lack of penetration will not appear on the radiograph; however, if lack of fusion (beyond the acceptance criteria) is present, lack of penetration will be suspect.

Note 2. Points A (dark) and B (light) will normally show as well defined geometric indications with Point A appearing as a dark linear indication and Point B as a light linear indication. The area that lies between the inside edges of indications from Points A & B (fit-up zone) is not to be evaluated for linear indications. All other type indications in the fit-up zones shall be evaluated. Other areas of the radiograph shall be evaluated as per standard practice.

Note 3. Utilize standard 3-AL-70R-3C for the evaluation of scatter porosity at the indicated radiographic positions.

Note 4. Special 2T Radiographic Qualification and mandatory Penetrameter/4T Visibility

(a) The procedure for sensitivity, 12007600A applies. Frequency of 2T qualifying radiograph to be one radiograph twice per week, Monday and Wednesday specifically. If at any time the contractor's radiographer/DCAS organization suspects loss of radiographic sensitivity, a radiographic qualification trial shot shall be taken as "prima facie" evidence of adequate 2T sensitivity.

(b) Penetrameters are to be used for the production radiographs for each radiographic position where the image of the penetrameter must be seen and the outline of the 4T hole visible before evaluating the radiograph for acceptance purposes.

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APPENDIX A

ALUMINUM WELDS

RADIOGRAPHIC MASTER INDEX
ACCEPTANCE LIST

RADIOGRAPHIC POSITION	DWG. NO.	SHEET NO.	DWG. ZONE
7903-1A	12007903	5	C-5
7903-1B	12007903	5	C-5
7903-2A	12007903	5	C-5
7903-2B	12007903	5	C-5
7903-3A	12007903	5	D-5,6
7903-3B	12007903	5	D-5,6
7903-4A	12007903	5	D-5,6
7903-4B	12007903	5	D-5,6
7924-1	12007924	1	E-3
7924-2	12007924	1	E-3
8101-1A	12008101	11	E-7
8101-1B	12008101	11	E-7
8101-1C	12008101	11	E-7
8101-1D	12008101	11	E-7
8101-2A	12008101	12	B-4,5
8101-2B	12008101	12	B-4,5
8101-2C	12008101	12	B-4,5
8101-2D	12008101	12	B-4,5
8101-3A	12008101	11	D-7

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APPENDIX A

ALUMINUM WELDS

RADIOGRAPHIC MASTER INDEX
ACCEPTANCE LIST (Cont'd)

RADIOGRAPHIC POSITION	DWG. NO.	SHEET NO.	DWG. ZONE
8101-3B	12008101	11	D-7
8101-4A	12008101	12	C-7
8101-4B	12008101	12	C-7
8101-5A	12008101	15	C-5, B-4 *
8101-5B	12008101	15	C-5, B-4 *
8101-6A	12008101	15	C-5, B-4 *
8101-6B	12008101	15	C-5, B-4 *
8101-17A	12008101	16	A-2
8101-17C	12008101	16	B-1, 2
8101-18A	12008101	16	A-2
8600-1	12008600	5	B-3
8600-2	12008600	5	C-3
8600-3	12008600	5	G-3
8600-4	12008600	5	H-3
8601-1	12008601	4	G-3
8601-2	12008601	4	H-4
8601-3	12008601	4	B-3
8601-4	12008601	4	C-4

* Process control radiographs which are not subject to Government acceptance/rejection decisions.

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APPENDIX A

ALUMINUM WELDS

RADIOGRAPHIC MASTER INDEX
ACCEPTANCE LIST (Cont'd)

RADIOGRAPHIC POSITION	DWG. NO.	SHEET NO.	DWG. ZONE
9107-1	12009107	5	B-3
9107-2	12009107	5	C-3
9107-3	12009107	5	G-3
9107-4	12009107	5	H-3
9108-1	12009108	4	G-3
9108-2	12009108	4	H-4
9108-3	12009108	4	B-3
9108-4	12009108	4	C-4

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APPENDIX A

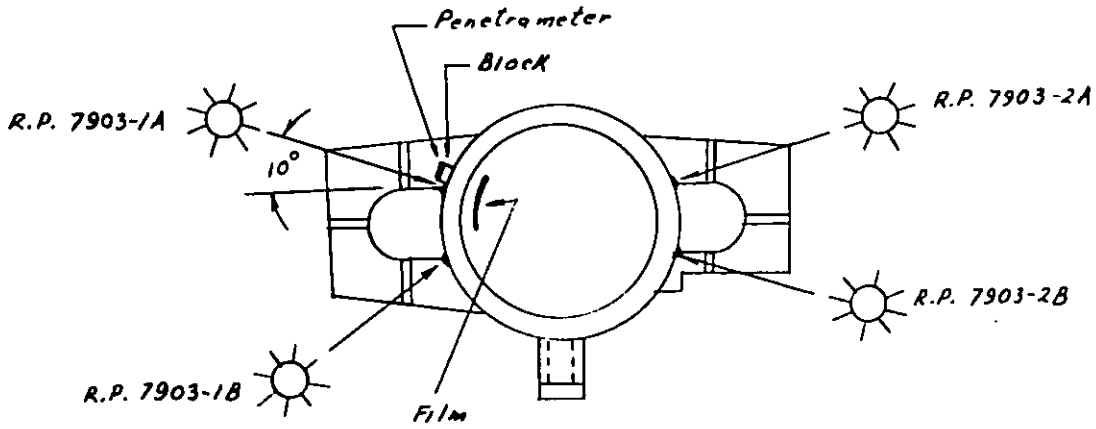
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 7903-1A, 1B, 2A, 2B

Part No.	1200-7903 Sh. 5 C-5	Part Name	Cradle
Film Type	Kodak M8 or equal	Film Size	5" x 7"
Penetrameter	1.5	Penetrameter Block	5/16"
Screen Pb Front	In Film Pack	Screen Pb Back	In Film Pack
K.V.	200 + 10%	Time at 3 M.A.	
Distance-source to film	30"	Time at 5 M.A.	1 Minute
Density Required	1.5 - 4.0	Time at 8 M.A.	

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - 1.5 inches



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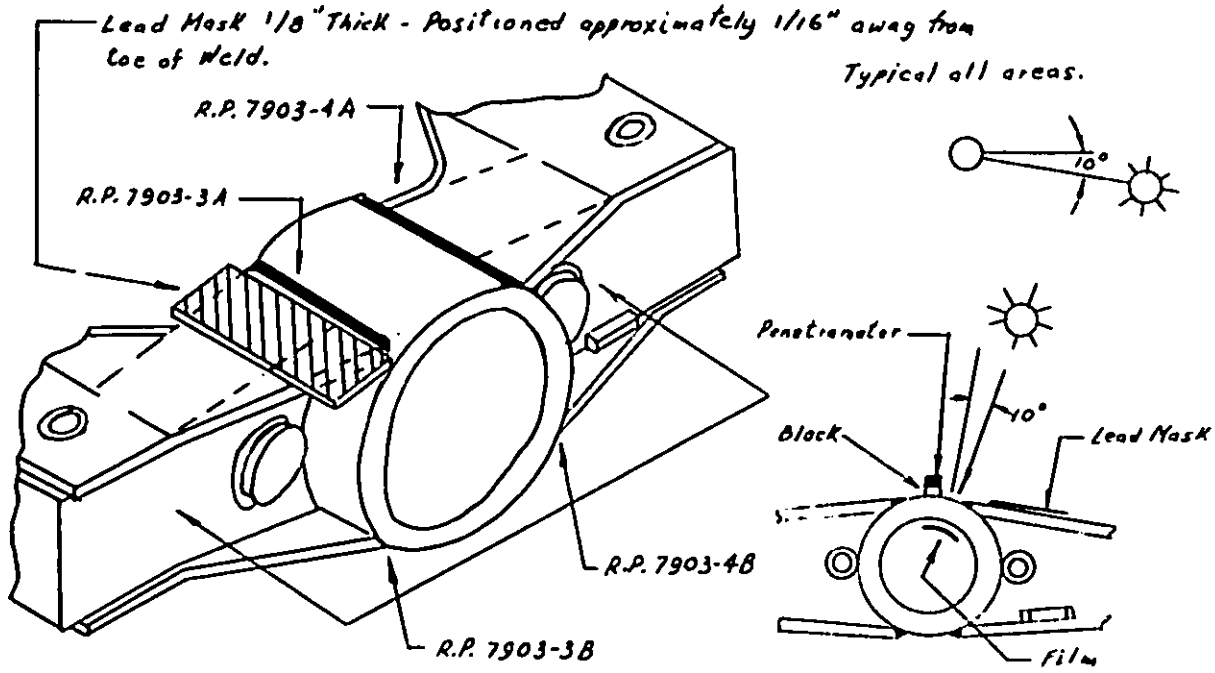
APPENDIX A

RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 7903-3A, 3B, 4A, 4B

Part No.	1200-7903 Sh. 5 E-6	Part Name	Cradle
Film Type	Kodak M8 or equal	Film Size	4 1/2" x 17"
Penetrameter	1.8	Penetrameter Block	7/16"
Screen Pb Front	In Film Back	Screen Pb Back	In Film Pack
K.V.	200 + 10%	Time at 3 M.A.	
Distance-source to film	30"	Time at 5 M.A.	1 Minute
Density Required	1.5 - 4.0	Time at 8 M.A.	

Acceptance Criteria RS 12007600 AL



Length of Radiation Path - 1.8 inches

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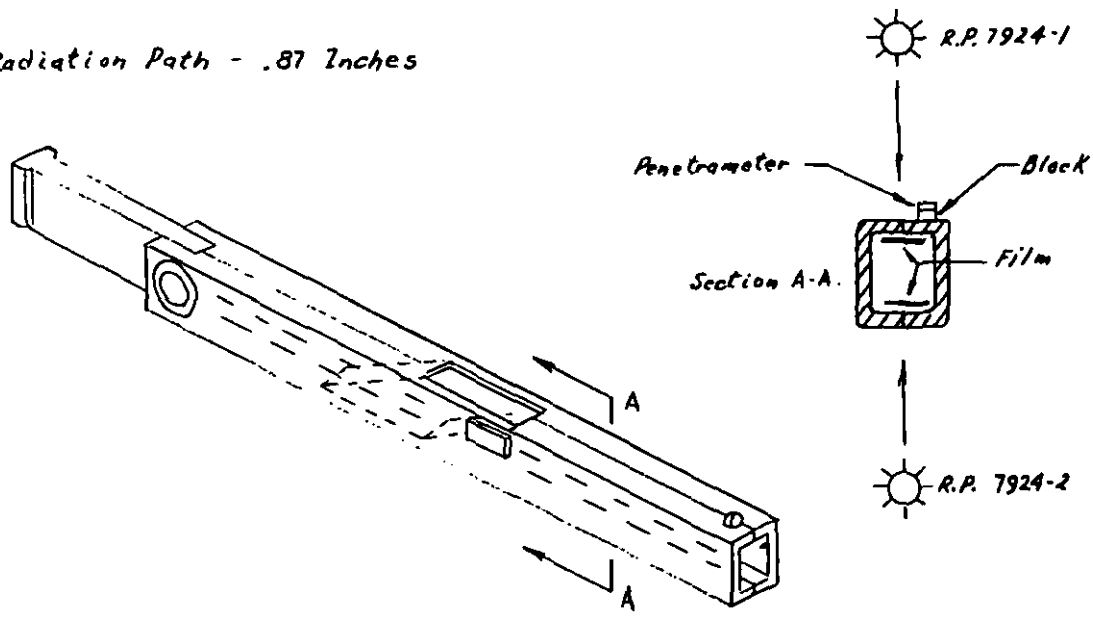
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 7924-1, 7924-2

Part No. 12007924 E5	Part Name Cradle Arm
Film Type Kodak M8 or equal	Film Size 4 1/2" x 17"
Penetrameter .87	Penetrameter Block 3/8"
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K.V. 145 + 10%	Time at 3 M.A.
Distance-source to film 30"	Time at 5 M.A. 1 1/2 Minutes
Density Required 1.5 - 4.0	Time at 8 M.A.

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - .87 Inches



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APPENDIX A

RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8101-1A, 1B, 1C, 1D, 2A, 2B, 2C, 2D

Part No. 1200-8101 Sh. 11 E-7
Sh. 12 B-4

Part Name Upper Carriage Weldment

Film Type Kodak M8 or equal

Film Size 4 1/2" x 17"

Penetrameter 2.0

Penetrameter Block 1/2"

Screen Pb Front In Film Pack

Screen Pb Back In Film Pack

K.V. 200 + 10%

Time at 3 M.A.

Distance-source to film 30"

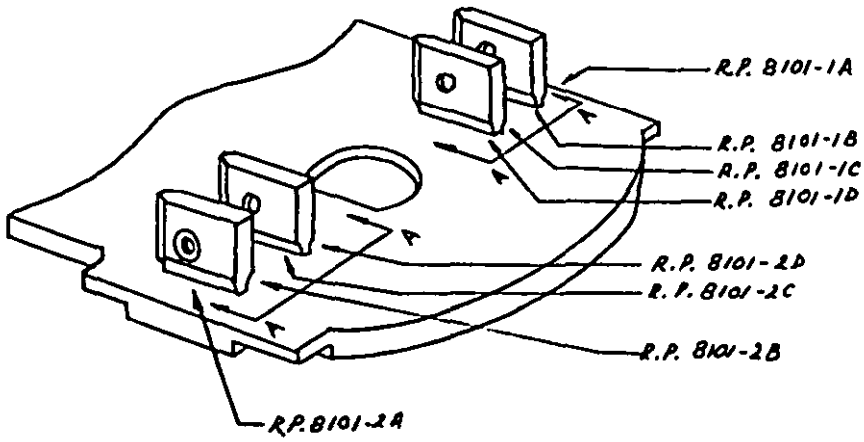
Time at 5 M.A. 2 Minutes

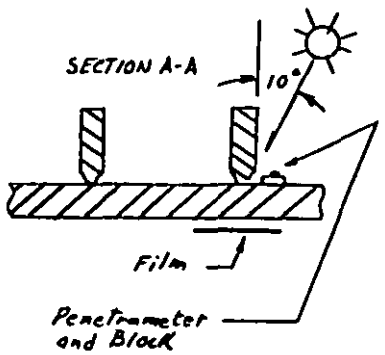
Density Required 1.5 - 4.0

Time at 8 M.A.

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - 2 Inches





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APPENDIX A

RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8101-3A, 3B, 4A, 4B

Part No 1200-8101 Sh. 11 C-7

S. 12 C-7

Part Name Upper Carriage Weldment

Film Type Kodak M-8 or equal

Film Size 4 1/2" x 17"

Penetrameter .37

Penetrameter Block 1/4"

Screen Pb Front In Film Pack

Screen Pb Back In Film Pack

K.V. 110 + 10%

Time at 3 M.A.

Distance-source to film 36"

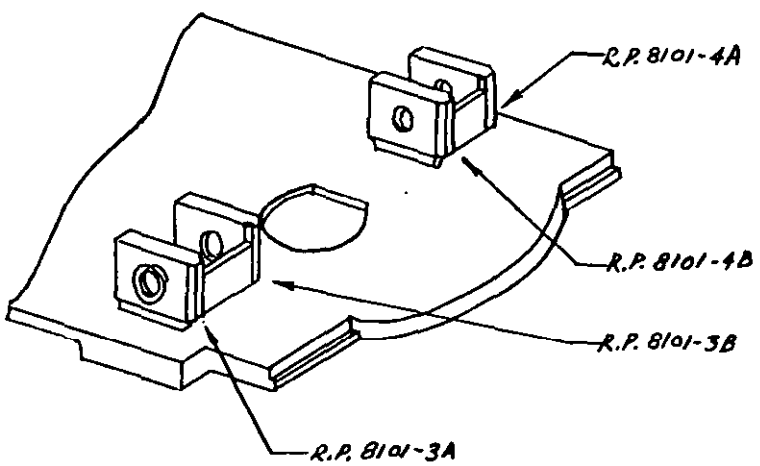
Time at 5 M.A. 4 1/2 Minutes

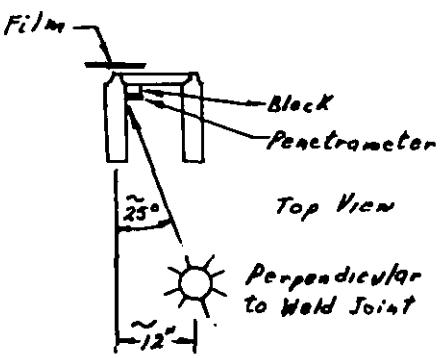
Density Required 1.5 - 4.0

Time at 8 M.A.

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - .37 Inches





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APPENDIX A

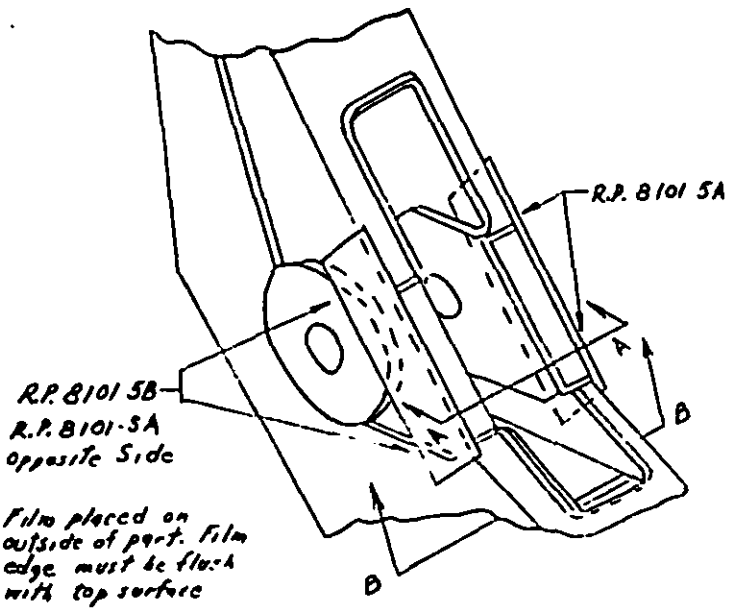
RADIOGRAPHIC TECHNIQUE DATA SHEET

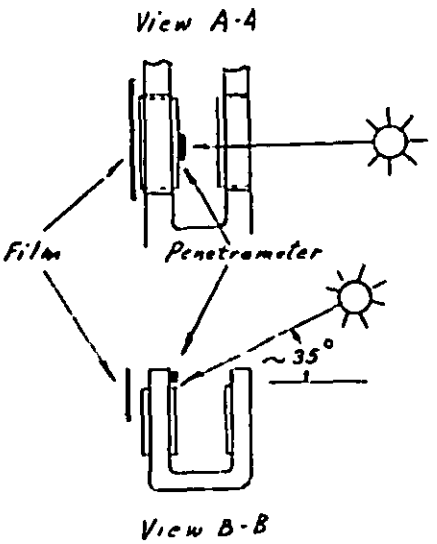
Radiographic Position No. 8101-5A and 5B

Part No. 1200-8101 Sh. 15 C-7	Part Name L/H Arm Sub Assy
Film Type Kodak M-8 or equal	Film Size 4 1/2" x 17"
Penetrameter 2"	Penetrameter Block None
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K.V. 200 + 10%	Time at 3 M.A.
Distance-source to film 30"	Time at 5 M.A. 2 Minutes
Density Required 1.5 - 4.0	Time at 8 M.A.

Acceptance Criteria Process Control Only

Length of Radiation Path - 2 Inches





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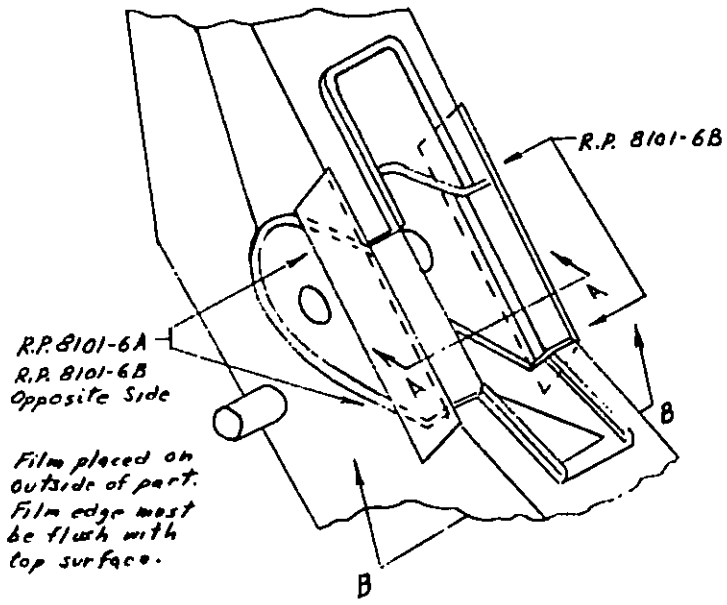
RADIOGRAPHIC TECHNIQUE DATA SHEET

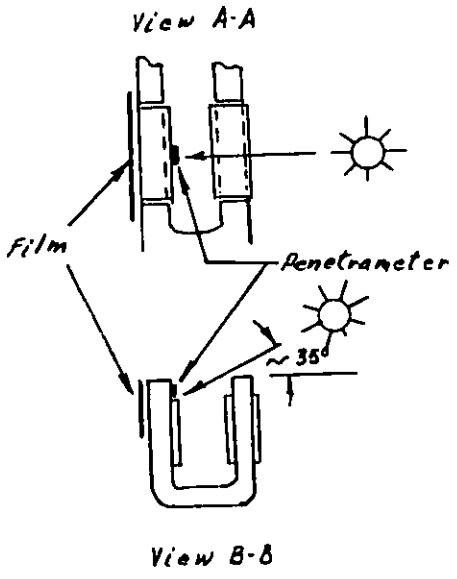
Radiographic Position No. 8101-6A and 6B

Part No.	12008101 Sh. 15 C-7	Part Name	R/H Arm Sub Assy
Film Type	Kodak M-8 or equal	Film Size	4 1/2" x 17"
Penetrameter	2"	Penetrameter Block	None
Screen Pb Front	In Film Pack	Screen Pb Back	In Film Pack
K.V.	200 + 10%	Time at 3 M.A.	
Distance-source to film	30"	Time at 5 M.A.	2 Minutes
Density Required	1.5 - 4.0	Time at 8 M.A.	

Acceptance Criteria Process Control Only

Length of Radiation Path - 2 Inches





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APPENDIX A

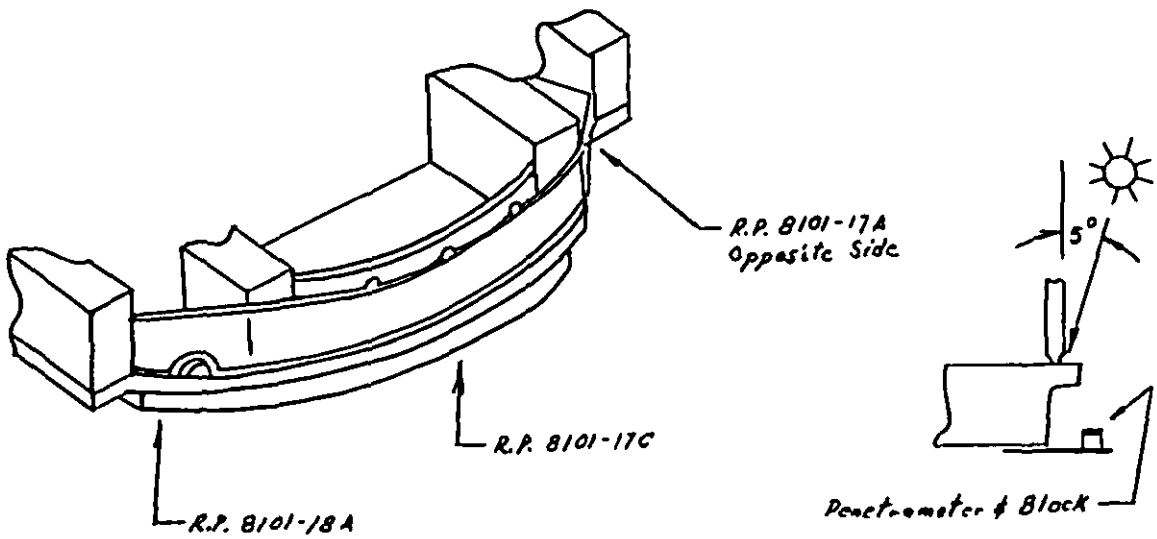
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8101-17A, 17C and 18A

Part No.	12008101 Sh 16 B-2	Part Name	Upper Carriage
Film Type	Kodak M-8 or equal	Film Size	5" x 7"
Penetrameter	.5	Penetrameter Block	1/2"
Screen Pb Front	In Film Pack	Screen Pb Back	In Film Pack
K.V.	110 + 10%	Time at 3 M.A.	
Distance-source to film	36"	Time at 5 M.A.	4 Minutes
Density Required	1.5 - 4.0	Time at 8 M.A.	

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - .38 Inches.



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APPENDIX A

RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8600-1, -1, -3, -4

Part No. 1200-8600 Sh. 5
Zones B-3, C-3, G-3, H-3

Part Name Trail R/H

Film Type Kodak M-8 or equal

Film Size 5" x 7"

Penetrameter 1"

Penetrameter Block 1/4"

Screen Pb Front In Film Pack

Screen Pb Back On Film Pack

K.V. 150 + 10%

Time at 3 M.A.

Distance-source to film 30"

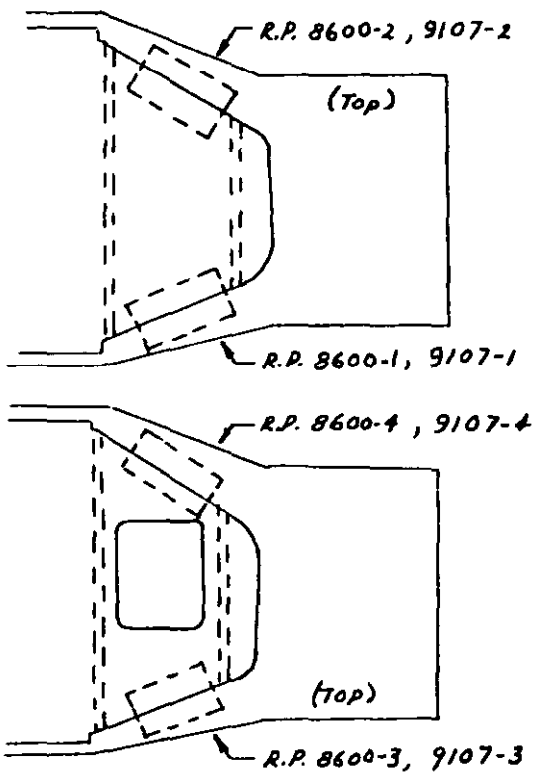
Time at 5 M.A. 1 Minute

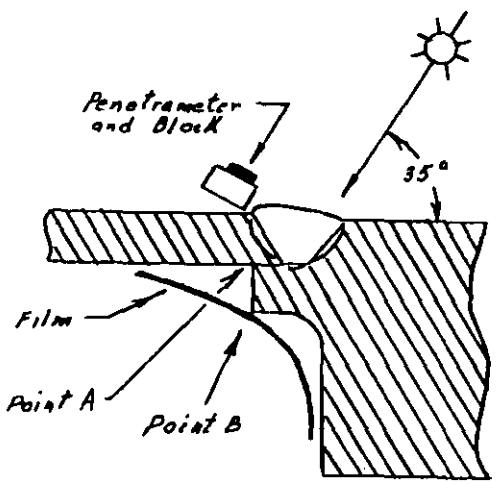
Density Required 1.5 - 4.0

Time at 8 M.A.

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - 1 Inch





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APPENDIX A

RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8601-1, -2, -3, -4

Part No. 1200-8601 Sh. 4

Zone B-4, C-4, G-4, H-4

Part Name Trail L/H

Film Type Kodak M-8 or equal

Film Size 5" x 7"

Penetrameter 1"

Penetrameter Block 1/4"

Screen Pb Front In Film Pack

Screen Pb Back In Film Pack

K.V. 150 + 10%

Time at 3 M.A.

Distance-source to film 30"

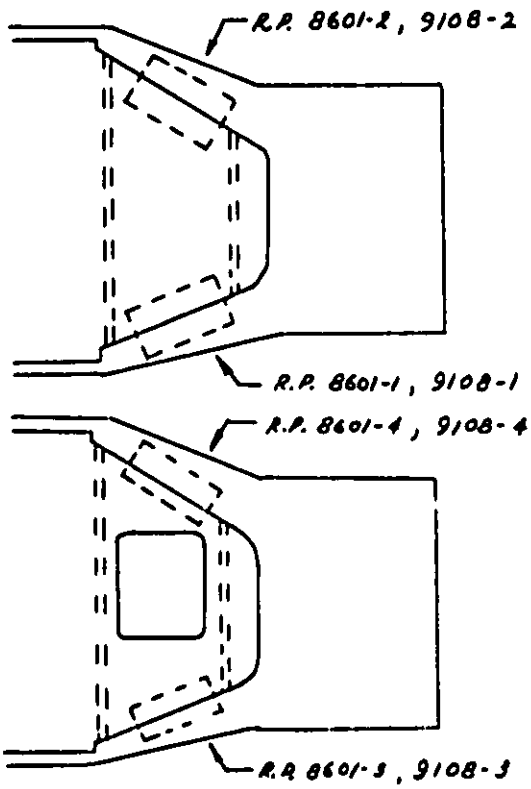
Time at 5 M.A. 1 Minute

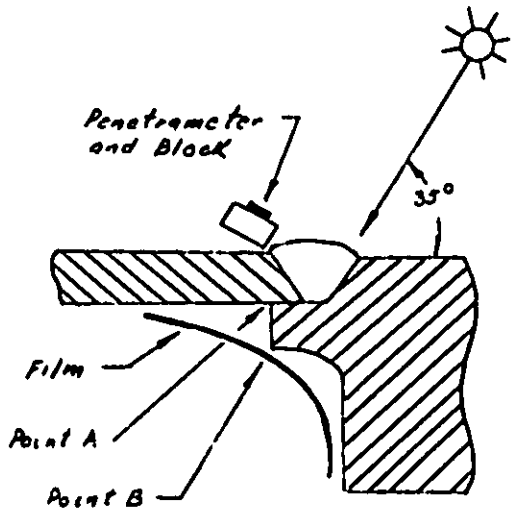
Density Required 1.5 - 4.0

Time at 8 M.A.

Acceptance Criteria RS 12007600 AL

Length of Radiation Path - 1 Inch





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APPENDIX A

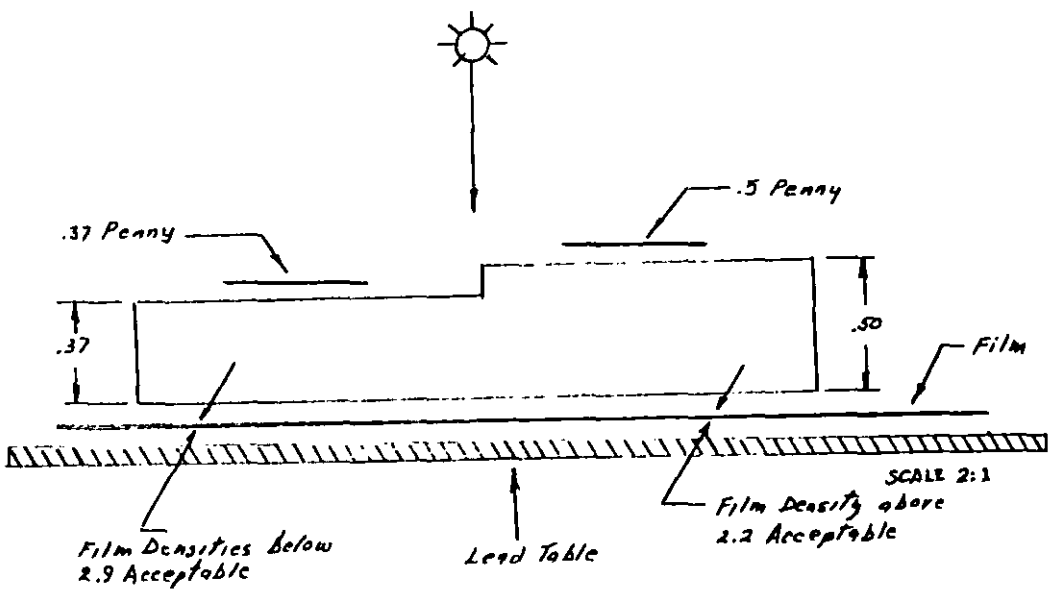
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 12007600A

Part No. 12008101	Part Name Top Carriage
Film Type Kodak M-8 or equal	Film Size 4 1/2 x 8 1/2
Penetrameter .37 and .5	Penetrameter Block Step Wedge - Below
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K.V. 115 to 130	Time at 3 M.A.
Distance-source to film 36"	Time at 5 M.A. 4.25 Minutes
Density Required 2.2 to 2.9	Time at 8 M.A.

Acceptance Criteria 2T Sensitivity

Identify Film with Exposure Number, Position No. 12007600A and Date.



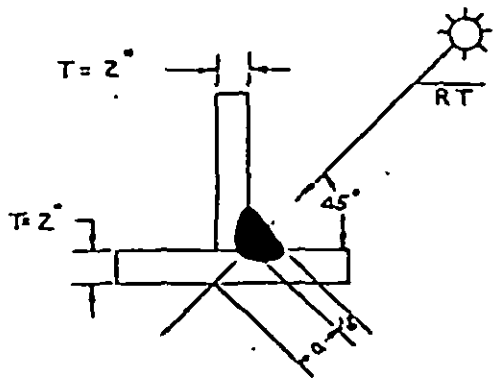
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APPENDIX A

RS 12007600AL

Aluminum Weldment

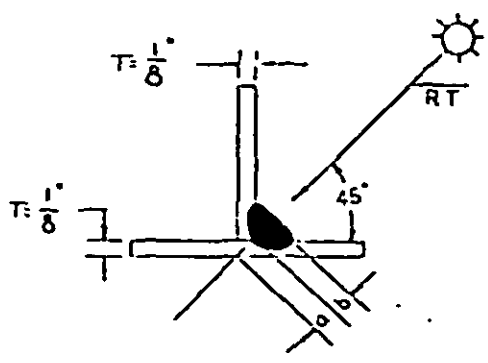
Example 1. Two inch (2") thick material welded to two inch (2") thick material. See Figure 1.



Let a = 2/sin 45° = 2/.707 = 2.82;
Let b = 3/8 inch.
Radiation Path = a + b = 2.82 + .375 = 3.203". Use rule para. 4.4.e.2 Weld and Parent joint material cross-section (radiation path) 1/2 inch and greater.

Rule 4.4.e.2: Since .3 x 2" = .6", then because .6" is greater than .0625" (1/16"), and 1/16" is the limit of defect width. Also, 1T in 12T or 1 (2") in 12 (2") or 1 inch of defect length in 12 inches of weld length of size no greater than .0625 inch width.

Example 2. One-eight (1/8") inch thick material welded to one-eight (1/8") inch thick material. See Figure 2.



Let a = 1/8 x 1/sin 45° = .1768 inch;
Let b = 1/4 inch.
Radiation Path = a + b = .1768 + .250 = .4268 inch. Use rule para. 4.4.e.1, Weld and Parent Joint materiel cross section (radiation path) less than 1/2".

Rule 4.4.e.1: Since defect width is 1/32" but 1/16 inch, indications in the root area shall be acceptable, then the maximum defect length allowed is 1/8" in every 1-1/2 inch of weld.

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APPENDIX B

RS 12007600 ST

10. SCOPE

10.1 This procedure provides standard reference radiographs for radiographic inspection of partial penetration weldments in steel plate, castings or forgings up to two (2) inches thick. They apply to full penetration welds also, as applicable.

10.2 Classification. The reference radiographs comprised in this standard are intended primarily for use on welds made with the coated stick electrode and inert gas metal-arc process.

20. APPLICABLE DOCUMENTS

SPECIFICATIONS

MILITARY

MIL-W-1261	-Welding Procedures for Constructional Steels

STANDARDS

MIL-STD-410	-Nondestructive Testing Personnel, Qualification and Certification
MIL-STD-00453 (USAF)	-Inspection, radiographic.
ATAC-STD-114	-Reference standards and radiographic procedures for partial penetration steel welds (Only to the extent specified herein.) Prepared by the U.S. Army Tank Automotive Command.
ATAC-STD-113	-Reference standards and radiographic procedures for partial penetration aluminum welds. (Only to the extent specified herein.) Prepared by the U.S. Army Tank Automotive Command.

30. DEFINITIONS

30.1 Partial Penetration Joint Design. The term partial penetration joint as used in this standard is defined as a weld joint containing an intentionally unfused area in the central portion of the joint.

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40. GENERAL REQUIREMENTS

40.1 Radiographic Inspection. Weldments shall be radiographically inspected in the locations specified on the applicable radiographic position chart drawings, radiographic technique data sheets, component drawings, specification, or as specified in the contract. Laboratories and personnel performing radiographic inspection shall meet the requirements of MIL-STD-00453 and MIL-STD-410, as applicable.

40.1.1 Radiographic Frequency. All steel weldments where the R.T. symbol is shown on the drawings shall be radiographed 100% unless drawing notes associated with the radiographic examination states otherwise.

40.2 Radiographic Variance. Because specialized techniques are required to facilitate radiographic inspection, it should be noted that some discontinuites may be masked due to the configuration of the specific joint design. For this reason, the joints will be inspected as follows:

40.3 Weld Designation: Inspection Categories.

1. Category 1 -Radiograph to be inspected for Scatter Porosity, Cluster Porosity, Slag Inclusions, Incomplete Fusion, Linear Porosity and cracks.

2. Category 2 -Radiograph to be inspected for Scatter Porosity, Cluster Porosity, Linear Porosity, Slag Inclusions Incomplete Fusion, Inadequate Penetration and cracks.

40.4 Inspection Acceptance Standard. "Reference Radiographs ATAC-STD-114, and ATAC-STD-113 apply to the extent specified below."

a. Scatter Porosity Coarse (fine & coarse) STD 2-FE-87-R2

b. Linear porosity STD 1-FE-114C-1

c. Cluster porosity, STD 1-FE-93L-1

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d. Slag Inclusions, STD 1 FE-48L-1

e. Gas Cavities: Shall not exceed 1/8" diameter with a pore frequency equal to STD1-AL45-1. (ATAC-STD-113).

f. Inadequate penetration and incomplete fusion:

1. Weld and Joint Material cross section (Radiation path) less than 1/2":

(a) 1/32" width or less rule - Indications in the root area shall be acceptable if the indications are less than or equal to a 1/32 inch in width, regardless of length in the root area.

(b) Greater than 1/32" but not greater than 1/16" in width, and length rule - Indications in the root area shall be acceptable if the indications are greater than 1/32 inch in width but not greater than 1/16 inch in width and the total length of the indications are not greater than 1T for any 12T length of weld (T is the thickness of the thinnest parent metal member of the weld joint).

(c) Greater than 1/16" width rule - Indications in the root area are not acceptable that are greater than 1/16 inches in width, regardless of length in the root area.

2. Weld and Joint Material cross section (Radiation path) 1/2 inch and greater:

(a) Thirty percent (30%) size and length rule - Indications shall be acceptable in the root area if they have a width not in excess of 30% of the thinnest parent metal member of the weld joint, (but not to exceed 1/16 inch in width)

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and
the total length of the indications in the root area are not greater than 1T for any 12T length of weld (T is the thickness of the thinnest parent metal member of the weld joint).

(b) Greater than 1/16" width rule - Indications in the root area are not acceptable that are greater than 1/16" in width, regardless of length, in the root area.

SEE LAST PAGE OF APPENDIX FOR EXAMPLES FOR THE INADEQUATE PENETRATION AND INCOMPLETE FUSION RULES.

g. Cracks are not acceptable.

h. Foreign Inclusions: All foreign material which have become an integral part of the weld shall be judged as follows:

1. Rounded indications shall be judged as per STD-2-FE-87-R2, STD-1-FE-114C-1 and STD-1-FE-93L-1.

2. Linear indications shall be judged as per the Slag Inclusions, STD-1-FE-48L-1.

40.4.1 Rounded indications of minor axis under 1/32 inch are not to be evaluated. Rounded indications have a major to minor axis dimensional ratio of three or less. They may be irregular in shape and may have tails which shall be included in evaluating the size.

40.4.2 Rejected welds shall be repaired and reinspected radiographically to the original standard.

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40.4.3 Various Flaw Types. With the exception of cracks, the common internal flaws experienced in steel weldments made with coated stick electrode and inert gas metal-arc process, are illustrated in the reference standards forming part of this procedure. All cracks are rejected.

a. Scattered porosity (fine and coarse). This flaw consists of scattered voids formed by gases failing to escape during weld metal solidification. On the radiograph, they are dark, round or elongated spots of varying size and density.

b. Clustered porosity. Clustered porosity appears the same as scattered porosity, except the pores are concentrated in one area, generally at the start of a weld bead or an interrupted arc.

c. Slag Inclusions. This flaw is generally derived from electrode coating materials or fluxes employed in welding operations. In multilayer welding operations, failure to remove the slag between layers will result in slag inclusions in these zones.

d. Linear porosity. The type most common in welds results from expanding gases generated in the normally unfused area. The cavities are linearily distributed at the root of the weld deposit and generally range from 1/32 to 1/16 inch diameter.

e. Gas Cavities. This flaw generally results from inadequate shielding gas or severe contamination of the gas. (It appears the same as scattered porosity except the average cavity is approximately 1/8 inch diameter.)

f. Inadequate penetration. In partial penetration and full penetration welds, this flaw consists of a linear root void in excess of the unfused area normally present. It appears on the radiograph as a straight dark line at either or both roots.

g. Incomplete fusion. Incomplete fusion is a failure of the weld metal to fuse completely with base metal or proceding bead. The condition resulting from trapped slag appears on the radiograph as a dark elongated line varying in width. The type not accompanied by slag generally occurs along the weld interface and produces a hair line unfused condition. This flaw is very difficult to detect radiographically due to its hairline condition and frequent interference from inadequate penetration indications.

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40.5 Departures from the inspection categories (paragraph 4.3) and the acceptance standard (paragraph 4.4) shall be as noted for the specific radiographic position in the Radiographic Classification Tables of this radiographic standard.

50. DETAILED REQUIREMENTS

50.1 Application of reference standards.

50.1.1 Determination of acceptability. Acceptability of production welds shall be determined by directly comparing the production radiograph with the designated reference standard for each type of flaw. The extent of flaw exhibited in the designated standard will be permitted throughout the length of any particular weld joint provided the flaw content in any weld length equal to that of the reference standard does not exceed that shown in the standard. Exceptions to this are required for inadequate penetration and incomplete fusion because acceptability of these flaws is based on material thickness.

50.1.2 Evaluation of single flaws. When the production radiograph shows flaws which are equal to or less severe than the designated acceptance standard specified for those types of flaws, the weld shall be considered radiographically acceptable. If the radiograph shows flaws of greater severity than specified, the weld shall be rejected unless repair is permitted.

50.1.3 Multiple flaws.

a. When the production radiograph shows one predominant type of flaw to an extent equal to the designated acceptance standard, and all other types to be less severe than the designated standard, the weld shall be considered radiographically acceptable.

b. When the production radiograph shows two types of flaws to an extent equal to the designated acceptance standard for each type of flaw, the weld may be accepted; however, an investigation shall be immediately instituted to correct the welding process or processes that resulted in the multiple flaws being equal to the standards. Subsequent radiographs of this weldment on the next succeeding unit displaying two types of flaws equal to the standard shall be rejected. A frequency of no more than one weldment in a total of twenty identical weldments fabricated shall be considered acceptable under this criteria.

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c. When the production radiograph shows three or more types of flaws to an extent equal to the designated acceptance standard for each type of flaw, the weld shall be rejected.

50.1.4 False indications. At times, radiographic films may indicate weld flaws when actually the film is defective. If doubt exists as to whether the flaw is in the weld or a film imperfection, the weld shall be radiographed again.

50.2 Inspection records. Radiographic reports and films shall be maintained and made available upon request to the designated government agency in accordance with MIL-STD-00453.

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APPENDIX B

RS 12007600 ST

RADIOGRAPHIC CLASSIFICATION FOR STEEL WELDMENTS

Radiographic Position No.	Category
8380-1	1 See Note 1 and Note 5
8380-2	1 See Note 2 and Note 5
8380-3	1 See Note 3 and Note 5
8380-5	Process Control Only
8380-6	Process Control Only
8380-7	2
8380-8	2
8380-9A	1 See Note 4 and Note 5
8380-9B	1 See Note 4 and Note 5
8380-10A	1 See Note 4 and Note 5
8380-10B	1 See Note 4 and Note 5

NOTES:

Note 1. Lack of fusion to be evaluated on thin plate P/N 12008389 only.

Note 2. Lack of fusion to be evaluated on thin plate P/N 12008390 only.

Note 3. Lack of fusion to be evaluated on thick plate P/N 12008484 only. Also, radiograph will be evaluated for weld joint fix-up tolerances (to be identified).

Note 4. Lack of fusion to be evaluated in conical plate (1/4 inch) P/N 12008543 only.

Note 5. Special 2T Radiographic Qualification & Mandatory Penetrameter/4T Visibility

(a) The procedure for sensitivity, 12007600S applies. Frequency of 2T qualifying radiograph to be one radiograph twice per week: Monday and Wednesday specifically. If at any time the contractor's radiographer/DCAS organization suspects loss of radiographic sensitivity, a radiographic qualification trial shot shall be taken as "prima facie" evidence of adequate 2T sensitivity.

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(b) Penetrameters are to be used for the production radiographs for each radiographic position where the image of the penetrameter must be seen and the outline of the 4T hole visible before evaluating the radiograph for acceptance purposes.

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APPENDIX B

STEEL WELDS

RADIOGRAPHIC MASTER INDEX
ACCEPTANCE LIST

RADIOGRAPHIC POSITION	DWG. NO.	SHEET NO.	DWG. ZONE
8380-1	12008380	5	G-11
8380-2	12008380	5	G-11
8380-3	12008380	5	G-10
8380-5	12008380	5	F-11 *
8380-6	12008380	5	F-11 *
8380-7	12008380	4	D-13
8380-8	12008380	4	D-13
8380-9A	12008380	4	C-12
8380-9B	12008380	4	C-12
8380-10A	12008380	4	C-12
8380-10B	12008380	4	C-12

* Process control radiographs which are not subject to Government acceptance/rejection decisions.

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APPENDIX B

RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8380-1 and 8380-2

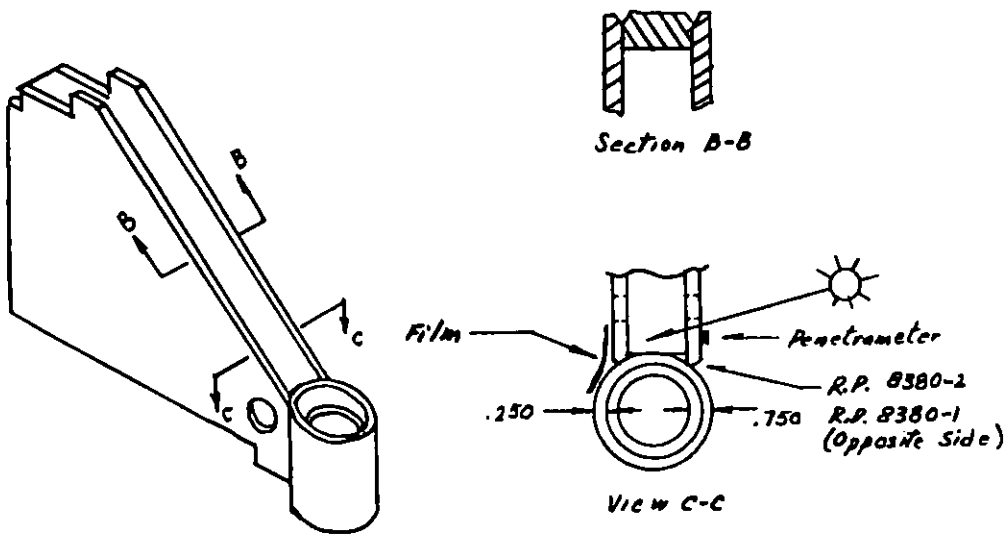
Part No. 1200-8380 Sh. 5 G-11	Part Name Bottom Carriage Weldment
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 8 (cut 4 1/2 x 17)
Penetrameter .62	Penetrameter Block NA
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K.V. 220 + 10%	Time at 3 M.A.
Distance-source to film 30"	Time at 5 M.A. 1 1/4 Minutes
Density Required 1.5 - 4.0	Time at 8 M.A.

Acceptance Criteria RS 12007600 ST

Length of Radiation Path - .62 Inches.

R.P. 8380-2 Shown

R.P. 8380-1 Opposite Side



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APPENDIX B

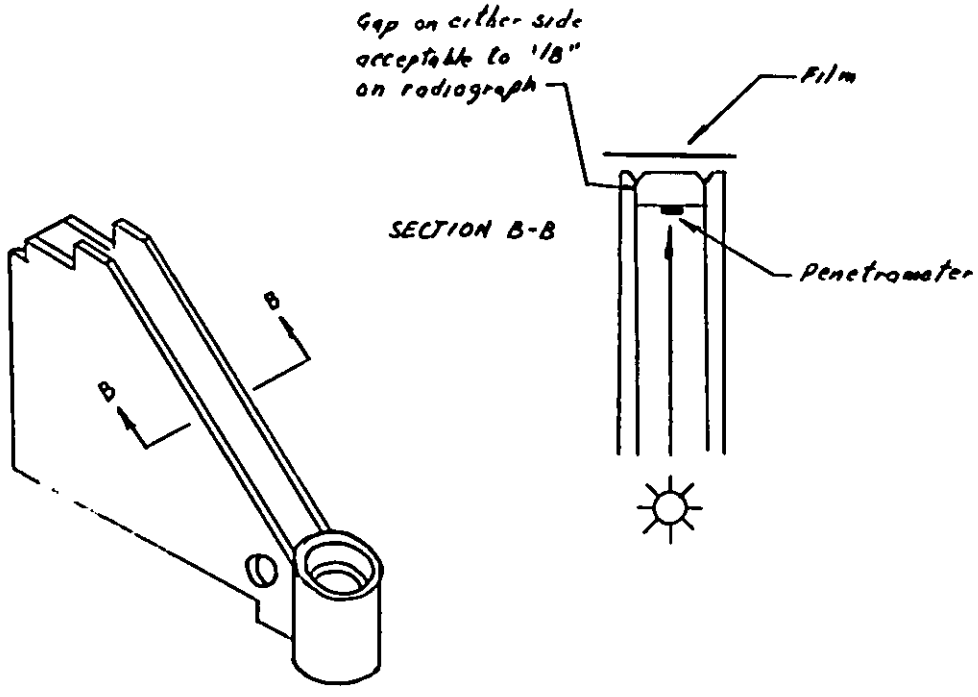
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8380-3

Part No.	1200-8380 Sh. 5 G-10	Part Number	Bottom Carriage Weldment
Film Type	Kodak AA-8 or equal	Film Size	4 1/2 x 17
Penetrameter	.50	Penetrameter Block	NA
Screen Pb Front	In Film Pack	Screen Pb Back	In Film Pack
K.V.	220 + 10%	Time at 3 M.A.	
Distance-source to film	30"	Time at 5 M.A.	1 1/4 Minute
Density Required	1.5 - 4.0	Time at 8 M.A.	

Acceptance Criteria RS 12007600 ST

Length of Radiation Path - .5"



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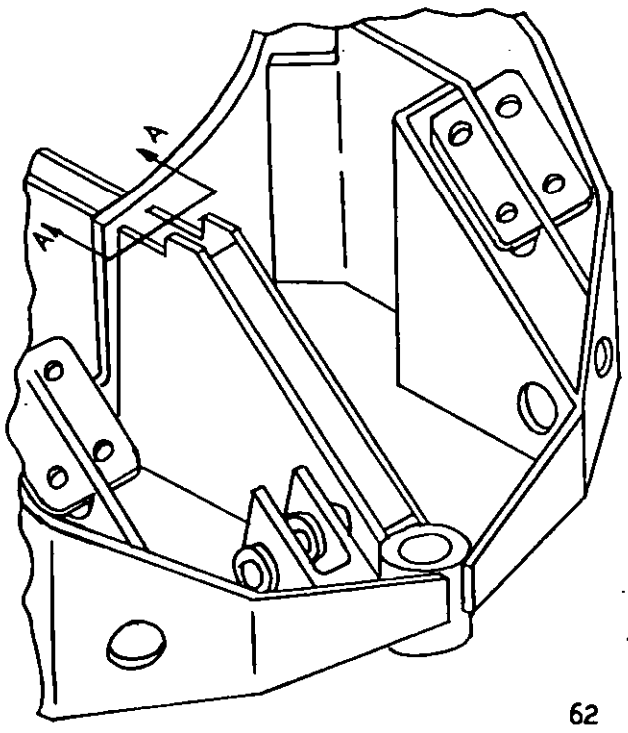
RADIOGRAPHIC TECHNIQUE DATA SHEET

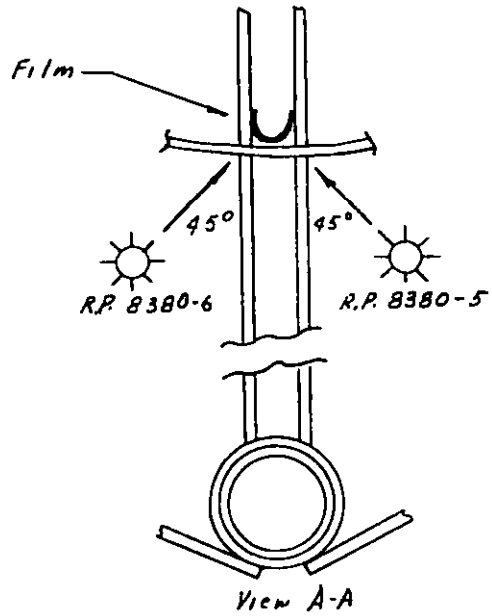
Radiographic Position No. 8380-5 and 8380-6

Part No. 1200-8380 Sh. 5 F-11	Part Name Bottom Carriage Weldment
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 17
Penetrameter .38	Penetrameter Block 3/8
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K. V. 220 + 10%	Time at 3 M. A. 1 1/4 Minutes
Distance-source to film 30"	Time at 5 M. A. 3/4 Minute
Density Required 1.5 - 4.0	Time at 8 M. A.

Acceptance Criteria Process Control Only

Length of Radiation Path - 3/8 Inches.





View A-A

NOTE: Put Penetrameter and Back-up Block on Portion of film which extends beyond area to be X-rayed.

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APPENDIX B

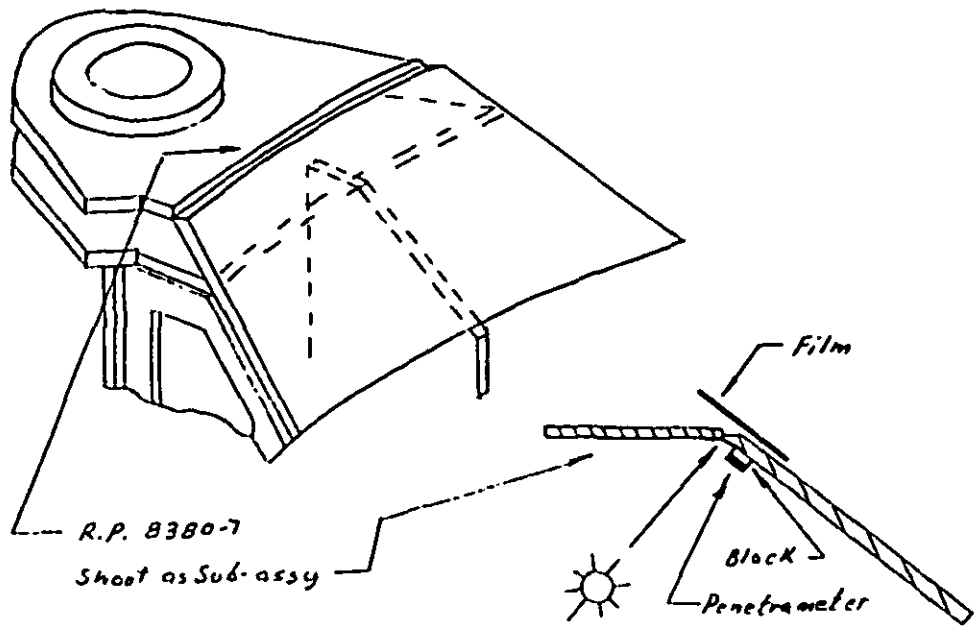
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8380-7

Part No. 1200-8380 Sh. 4 D-13	Part Name Bottom Carriage Weldment
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 17
Penetrameter .38	Penetrameter Block 1/16"
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K. V. 160 + 10%	Time at 3 M. A.
Distance-source to film 30"	Time at 5 M. A. 2 Minutes
Density Required 1.5 - 4.0	Time at 8 M. A.

Acceptance Criteria RS 12007600 ST

Length of Radiation Path - .38 Inches.



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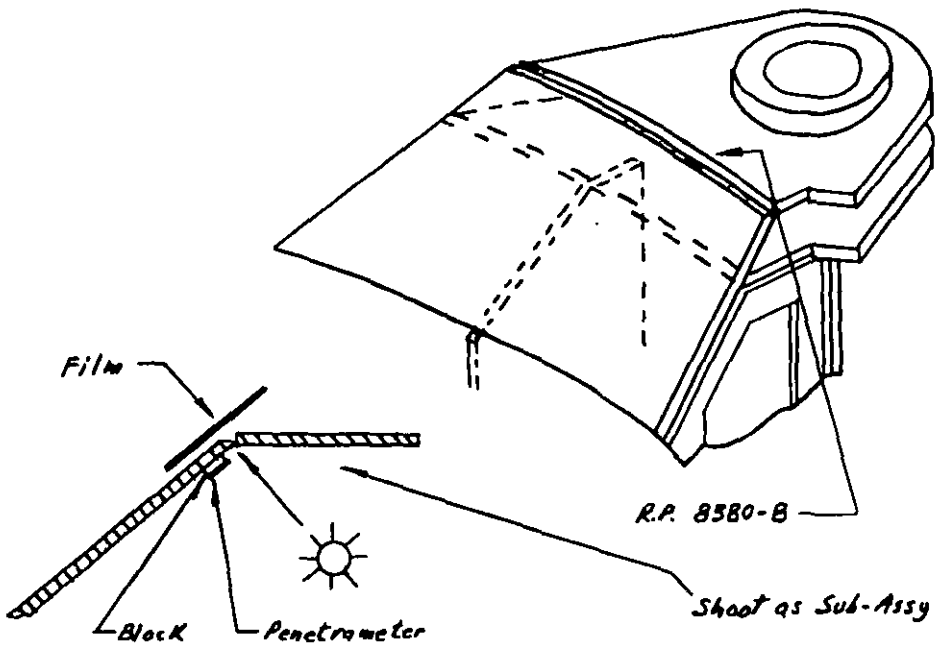
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8380-8

Part No. 1200-8380 Sh. 4 D-13	Part Name Bottom Carriage Weldment
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 17
Penetrameter .38	Penetrameter Block 1/16"
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K. V. 160 + 10%	Time at 3 M. A.
Distance-source to film 30"	Time at 5 M. A. 2 Minutes
Density Required 1.5 - 4.0	Time at 8 M. A.

Acceptance Criteria RS 12007600 ST

Length of Radiation Path - .38 Inches.



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APPENDIX B

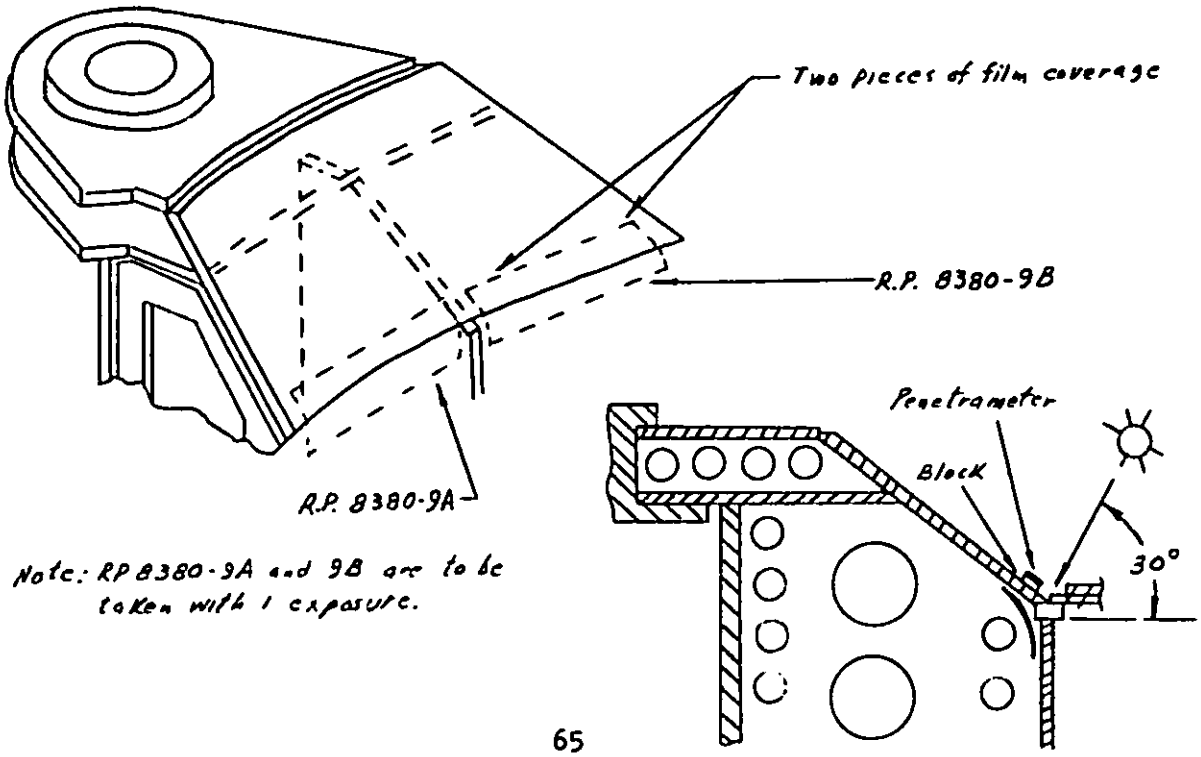
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8380-9A and 8380-9B

Part No. 1200-8380 Sh. 4 C-12	Part Name Bottom Carriage Weldment
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 17(9B) 4 1/2 x 8(9A)
Penetrameter .38	Penetrameter Block 1/8"
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K. V. 220 + 10%	Time at 3 M. A.
Distance-source to film 30"	Time at 5 M. A. 1 Minute
Density Required 1.5 - 4.0	Time at 8 M. A.

Acceptance Criteria RS 12007600 ST (LEFT HAND SIDE)

Length of Radiation Path - .38 Inches.



Note: RP 8380-9A and 9B are to be taken with 1 exposure.

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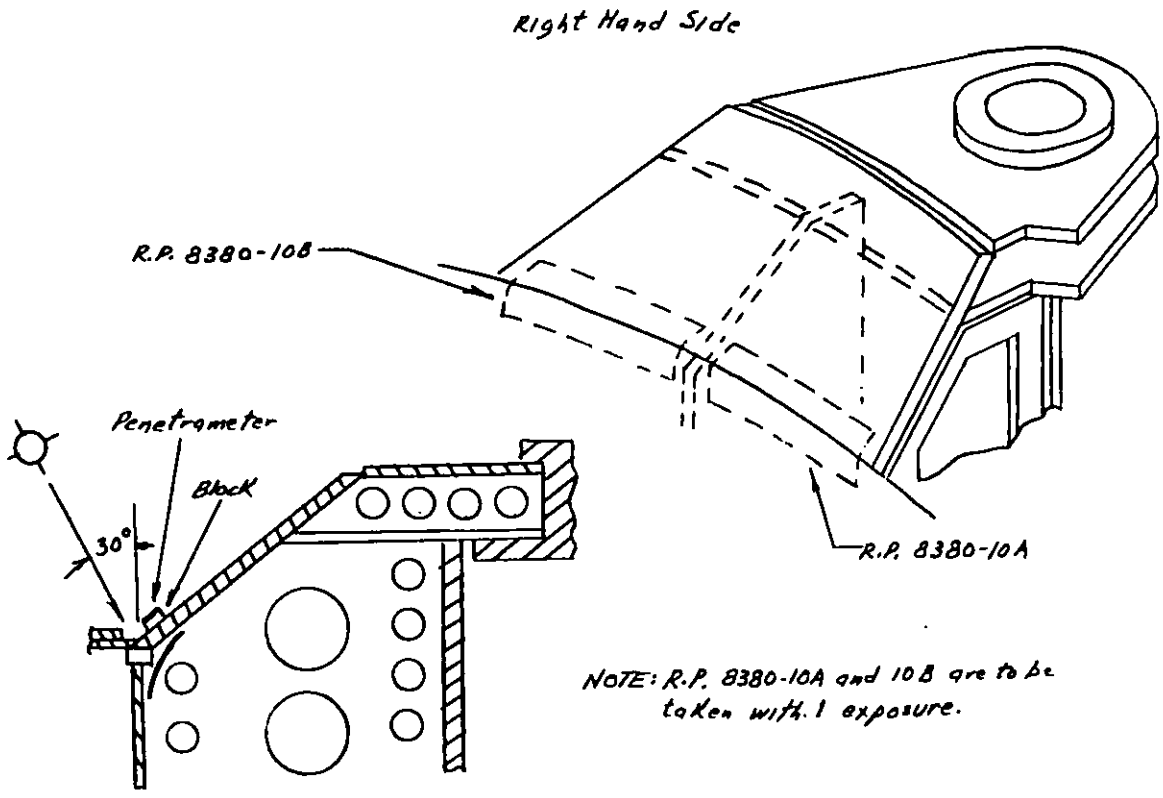
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 8380-10A and 8380-10B

Part No. 1200-8380 Sh. 4 C-12	Part Name Bottom Carriage Weldment
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 17(10B) 4 1/2 x 8(10A)
Penetrameter .38	Penetrameter Block 1/8"
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K. V. 220 + 10%	Time at 3 M. A.
Distance-source to film 30"	Time at 5 M. A. 1 Minute
Density Required 1.5 - 4.0	Time at 8 M. A.

Acceptance Criteria RS 12007600 ST

Length of Radiation Path - .38 Inches.



NOTE: R.P. 8380-10A and 10B are to be taken with 1 exposure.

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APPENDIX B

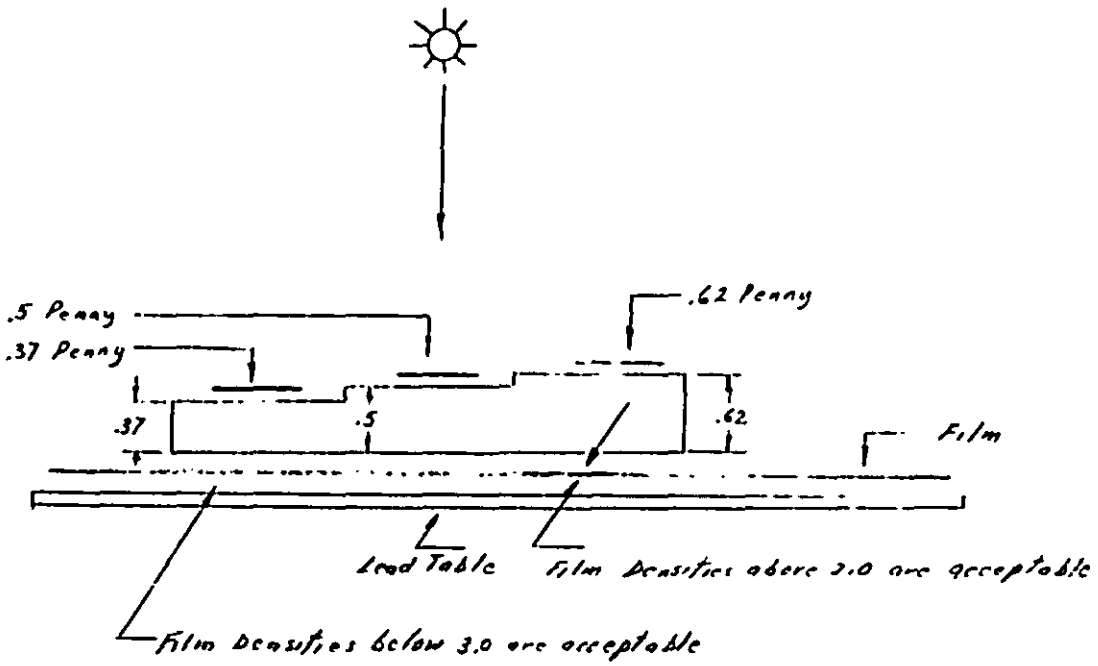
RADIOGRAPHIC TECHNIQUE DATA SHEET

Radiographic Position No. 12007600S

Part No. 1200-8380	Part Name Bottom Carriage
Film Type Kodak AA-8 or equal	Film Size 4 1/2 x 8 1/2
Penetrameter .37, .5 and .62	Penetrameter Block Step Wedge - Below
Screen Pb Front In Film Pack	Screen Pb Back In Film Pack
K. V. 205 to 235	Time at 3 M. A.
Distance-source to film 30"	Time at 5 M. A. 1.25 Minutes
Density Required 2.0 to 3.0	Time at 8 M. A.

Acceptance Criteria 2T Sensitivity

Identify Film with Exposure No., Position No. 12007600S and Date.



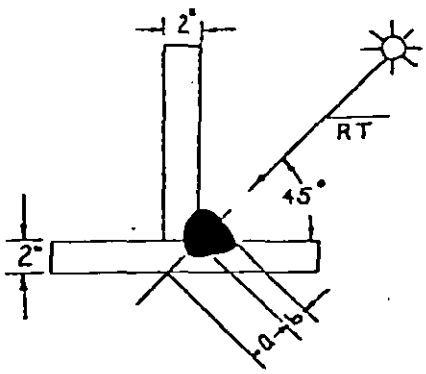
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APPENDIX B

RS 12007600ST

Steel Weldment

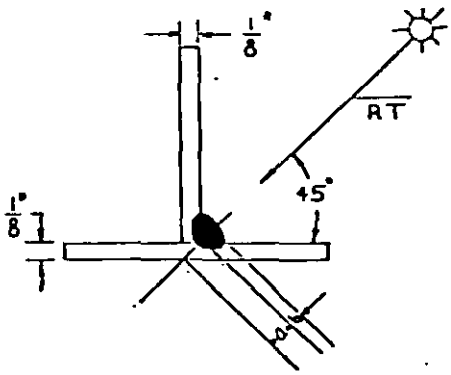
Example 1. Two inch (2") thick material welded to two inch (2") thick material. See Figure 1.



Let a = 2/sin 45° = 2/.707 = 2.82;
Let b = 3/8 inch.
Radiation Path = a + b = 2.82 + .375 = 3.203". Use rule para. 4.4.f.2 Weld and Parent joint material cross-section (radiation path) 1/2 inch and greater.

Rule 4.4.f.2: Since .3 x 2" = .6", then because .6" is greater than .0625" (1/16"), and 1/16" is the limit of defect width. Also, 1T in 12T or 1 (2") in 12 (2") or 1 inch of defect length in 12 inches of weld length of size no greater than .0625 inch width.

Example 2. One-eight (1/8") inch thick material welded to one-eight (1/8") inch thick material. See Figure 2.



Let a = 1/8 x 1/sin 45° = .1768 inch;
Let b = 1/4 inch.
Radiation Path = a + b = .1768 + .250 = .4268 inch. Use rule para. 4.4.f.1, Weld and Parent Joint materiel cross section (radiation path) less than 1/2".

Rule 4.4.f.1: Since defect width is 1/32" but 1/16 inch, indications in the root area shall be acceptable, then the maximum defect length allowed is 1/8" in every 1-1/2 inch of weld.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-45987A	CARRIAGE, 155MM, HOWITZER: M39

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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